

Abstract

Incremental Sheet Metal Forming (ISMF) is regarded as a cost-effective method for manufacturing sheet metals due to its elimination of the need for dedicated dies, which are typically required in conventional processes. This method, which utilizes CNC machines, offers high flexibility, allowing operators to easily modify product geometry, unlike methods such as stamping. This project evaluates and controls the ISMF process to achieve optimal performance. Specifically, it focuses on the experimental and Finite Element analysis of the effects of sheet thickness in Single Point Incremental Forming (SPIF) of Aluminum 1050. Initially, a series of forming tests were conducted with various forming parameters. The data from these experiments were used to analyze and understand the phenomena of sheet thinning generated during the forming process. The results also revealed the relationship between incremental depth and sheet thickness at different feed rates and rotation speeds for the configuration head. Additionally, the impact of different rotation speeds on sheet thinning at various levels was examined. Finite Element Analysis (FEA) was employed to simulate the SPIF process for Aluminum 1050 under specific process parameters. The simulations indicated that the minimum sheet thickness of Aluminum 6061 reached 23.9% of the original thickness. The study concluded with an optimization analysis to identify the optimal forming conditions.

Acknowledgments

The authors would like to give thanks to Allah the almighty, without whom this work could not have been completed.

The authors would like to extend special thanks and gratitude to all staff members of the Production Engineering & Mechanical Design Department, Port-Said University.

The authors gratefully acknowledge the support of **Eng. Ali Abdel aziz** and **Eng. Mohamed Moustafa** from the Suez Canal Authority workshop for his support in conducting the machining experiments.

The authors would like to thank **Eng. Mohamed Kouta** for helping with the CNC machine.

The authors would like to extend sincere thanks to **Eng. Moataz Abd Elkafy** for helping in optimization and **Eng. Khaled Mansour** .

Table of Contents

Abstract	i
Acknowledgments	iv
Table of Contents.....	vi
List of Figures	ix
CHAPTER 1: INTRODUCTION	5
1.1 Over view	5
1.2 Background for Incremental Sheet Metal Forming (ISMF)	
1.3 Aim of the work	
1.4 project outlines	
CHAPTER 2: LITERATURE REVIEV.....	8
2.1. Introduction to Metal Forming Processes.....	8
2.2 Types of Sheet Metal Forming Processes	8
2.2.1 Shearing.....	10
2.2.2 Bending.....	11
2.2.3 Deep Drawing.....	11
2.2.4 Punching	11
2.2.5 Embossing.....	11
2.2.6 Stretching.....	12
2.2.7 Rolling	12
2.2.8 Welding	12
2.2.9 Coating.....	12
2.2.10 Laser Cutting	12

2.3. Incremental Sheet Metal Forming (ISF)	13
CHAPTER 3: EXPERIMENTAL WORK.....	18
3.1 Overview.....	18
3.2 workpiece material	18
3.3 setup design.....	18
3.4 forming tool	18
3.5 sample deformation.....	19
3.6 forming parameters	20
3.7 Forming processing	21
3.8 FEM Analysis.....	23
Chapter 4 : RESULTS AND DISCUSSIONS	
4.1 Overview.....	26
4.2 Effect of incremental sheet metal forming parameters on surface roughness and sheet thickness.....	28
4.3 Effect of stress on sheet metal for 9 specimens (von mises):	30..
4.4 effect of plastics strain on sheet metals for 9 specimens :	35
4.5 effect of shell thickness on sheet metals for 9 specimens:	40
4.6 effect of displacement on sheet metals for 9 specimens:	42
Chapter 5 : conclusion	45
Refrences.....	47

CHAPTER 1

INTRODUCTION

1.1 Overview

Until recently, CNC machine tools have been primarily recognized for their role in metal cutting processes. However, CNC machines are actually designed to meet a wide range of design and manufacturing requirements for complex products. These machines possess the capability to significantly alter the nature of the manufacturing process, allowing for greater precision and flexibility in product execution. By utilizing advanced CNC tooling, manufacturers can achieve intricate designs and high-quality finishes that were previously difficult or impossible to attain with traditional methods. This adaptability makes CNC machines essential for modern production environments, where the demand for detailed and customized products is continually increasing.

Following the Second World War, the rapid advancement of aerospace products such as launch vehicles, satellites, and both military and civil aircraft presented significant manufacturing challenges. The electrical program-controlled machines and hydraulic copying methods available at the time were inadequate for the complex designs required by the aerospace industry. Engineers urgently needed to find a solution to automate these processes. To address the critical demands of the aerospace sector, numerical control (NC) technology was developed in the early 1950s. This innovation revolutionized manufacturing capabilities and laid the groundwork for modern automated machining processes. Many components used in aircraft and space vehicles are machined from solid raw materials, often implicating removal of considerable stock and requiring multiple hundreds of different positioning movements of the machine tool slides. The integration of computers with NC machine tools resulted in the following advantages; Increase in the memory for part program processing, Increase in capacity for storing large part programs. Easy editing of part programs on the control console, Realization of control logic through software and Significant improvement in the reliability of operation of the machines, NC machines integration in the manufacturing system as a whole(1, 2).

CNC machine tools integrated with CAD/CAM systems create a harmonious package that addresses numerous challenges in modern manufacturing. These CNC tools enable the rapid production of complex and difficult-to-manufacture parts. Additionally, CNC machines can perform processes such as locally deforming sheet metals using various tools.

The operation principle of a numerically controlled machine can be illustrated with the example of a 3D milling machine. For an operation like end milling, the spindle head must be positioned along the Z-axis, while the table is positioned along the X and Y coordinate axes. The feed movement is achieved through simultaneous or individual positioning of the X and Y axes. Thus, the milling machine requires three slide movements, commonly referred to as axis feed drives. To perform the milling operation on the workpiece, the coordinate information (X and Y coordinates) of the starting and ending points must be encoded in the NC program.

Amongst the materials which have been used in sheet metal forming Aluminum, steel, stainless steel and magnesium alloys are being used in the industry. Aluminum has one third the density of steel, however with such low density it possesses high strength and would decrease the weight of the structure when compared with steel

. The main elements of aluminum 6xxx series are magnesium and silicon are main elements. The 6xxx aluminum alloys have great corrosion resistance and high extrude-ability. Their ultimate tensile strength, modulus of elasticity and tensile yield strength are about 125-400MPa, 68.9 GPA and 276 MPa respectively .Formability and weld ability for these series are high. Moreover, they are used in automotive, exterior panels and marine industries as well as construction applications(3).

1.2 Background for Incremental Sheet Metal Forming (ISMF)

Sheet metal forming operations play an important role in producing varieties of products for a wide range of engineering applications(4). In sheet-metal operations, parts are processed by forming the material either in cold or hot conditions. In particular, when the workpiece material heated, it has a lower-resistance to deformation and thus many sheet metal parts are formed in hot condition(5).

Examples of sheet metal techniques include spinning, stamping, flow forming, hammering and Incremental Sheet Metal Forming (ISMF)(6). ISMF is an advanced method of metal forming processes recently emerged using computerized numerical control (CNC) potential. In particular, with the aid of CNC technology complex shapes(7). can be produced directly without using dies, by simply deforming tool with step by step incremental feed(7, 8). ISMF can be classified according to various points of view such as: the forming method, the part geometry, the forming path, tool path strategy, and the applied tools, etc. (9). The most common classification is based on the forming method and is divided to single-point and two-point. Single-point incremental forming (SPIF) can also called negative incremental forming, while Two-point incremental forming (TPIF) is also called positive incremental forming.

In sheet incremental forming, the sheet is precisely clamped and usually deformed by using a hemispherical tool following the required shape in space by a CNC controlled incremental tool movement as shown in Fig.1.1(10).

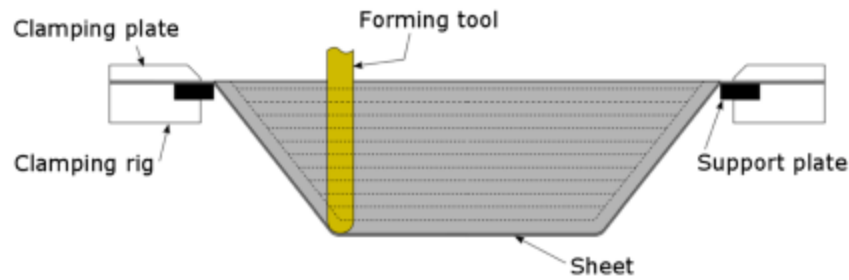


Figure 1.1 Basic principle of incremental forming(10)

Rigid blank holders are used to clamp the sheet. Often these generic clamping fixtures allow incorporation of different (dedicated) support or backing plates that help to limit undesirable deformation of the blank in an early stage of the forming process. Rigid support in the immediate vicinity of the forming zone assures a higher accuracy and reduces scrap material associated with extended transition areas between the actual part geometry and clamped workpiece edge. The sheet is typically clamped on the blank holder using bolts and clamping plates along the edge of the sheet. Finite element simulations by Bouffieux et al(11). have demonstrated that limited sliding can significantly affect the tool force required to form an indent shape. A sliding effect of 0.08 mm was, for example, found to result in a force reduction on the tool with 16%. In order to obtain reproducible results, the torque applied to each clamping bolt therefore has to be constant and sufficiently high to prevent sliding(10).

1.3 Aim of the work

The objective of this study is to systematically investigate the impact of various process parameters, including feed rate, tool rotational speed, incremental depth, etc., on the performance of the manufacturing process. This assessment focuses on evaluating dimensional accuracy, stress distribution, plastic strain of the final product shapes.

In order to fulfill the overall aim of the work, the following objectives were set:

1. Studying the relationship between incremental depth and sheet thickness at different feed rates.
2. Studying the relationship between incremental depth and sheet thickness at different speeds.
3. Studying the capability of the FEM process and the parameters that leads to better results.
4. figuring out the relationship between the spindle speed and the sheet thinning at different height.

1.4 project outlines

Following to Chapter 1 that introduces to this project, the remainder of the project is organized as follows:

Chapter (2): Literature Review

This chapter reports on the most relevant topic and discusses the previous related work to the research points intending to highlight the gaps of the current research and thus to be able to identify the lack of scientific understanding of the process.

Chapter (3): FEM Validation

This chapter outlines the experimental work including material properties used for experimental tests, forming tool specifications, CHAPTER 1 INTRODUCTION eleven forming conditions and experimental

procedure. In additions, the chapter discusses the measurement technique and devices for evaluating sheet thickness and surface roughness. Also, the chapter discusses the finite element method (FEM) and the modeling steps for simulating the SPIF process.

Chapter (4): Results and Discussions

This chapter extends detailed discussion of the theoretical and finite element results obtained. However, more attention is paid to explain the rationale behind the main findings.

CHAPTER 2 LITERATURE REVIEW

2.1. Introduction to Metal Forming Processes

The metal forming industry is vital in modern manufacturing, producing millions of tons of metals annually for various sectors. It embraces new technologies like automation and AI to enhance efficiency and quality. Research and development efforts lead to innovations in alloys, forming techniques, and eco-friendly practices. Sustainability focuses on waste reduction and energy efficiency through recycling and renewable sources. Workforce training ensures updated skills for safety and productivity. The industry is dynamic, adapting to diverse needs while addressing sustainability and workforce development for industrial advancement and economic prosperity.

2.2 Types of Sheet Metal Forming Processes

Sheet metal forming is the process of transforming flat metal sheets into various shapes. This is achieved through a variety of techniques, each suited to specific applications and desired outcomes. Here are some of the most common sheet metal forming processes:

1. Shearing: This process involves cutting sheet metal into desired shapes using specialized shears or dies. It's a fundamental operation in sheet metal fabrication, often used to prepare material for further processing

2. Bending: Bending is the process of folding sheet metal along a straight axis. This is typically done using a press brake, which applies force to bend the metal around a die. Bending is used to create angles, channels, and other shapes.

3. Deep Drawing: This process involves drawing a flat sheet of metal into a die cavity to create hollow shapes like cups, cans, and containers. It's a complex process that requires careful control of material flow and pressure.

4. Punching: Punching is used to create holes in sheet metal. A punch press forces a shaped punch through the metal, creating a clean hole. Punching is often used in conjunction with other processes like bending and shearing.

5. Embossing: Embossing is the process of creating raised or recessed designs on sheet metal. This is done by pressing the metal between a male and female die, which have the desired pattern. Embossing is often used for decorative purposes.

6. Stretching: Stretching involves applying tension to a sheet of metal to change its shape or increase its length. This is often used to create complex curved shapes or to form parts with tight tolerances.

7. Rolling: Rolling is the process of reducing the thickness of sheet metal by passing it between rollers. This can also be used to create curved shapes or to improve the surface finish of the metal.

8. Welding: Welding is the process of joining two or more pieces of sheet metal together using heat or pressure. There are many different welding techniques, each suited to specific materials and applications.

9. Coating: Coating involves applying a layer of material to sheet metal to protect it from corrosion or to improve its appearance. Common coatings include paint, powder coating, and zinc plating.

10. Laser Cutting: Laser cutting is a precise method of cutting sheet metal using a focused laser beam. It's often used for complex shapes or for applications where high accuracy is required.

2.3. Incremental Sheet Metal Forming (ISF)

Incremental Sheet Forming (ISF) is a sheet metal forming process that allows the manufacturing of components without the need for dedicated tools like those used in stamping. The final part geometry is largely influenced by the forming tool path, which also affects the distribution of sheet thickness.

Types of Incremental Sheet Forming

ISF is categorized into two main types:

- 1. Conventional Incremental Sheet Forming (CISF):** In CISF, a sheet of metal is progressively deformed using a simple hemispherical or ballpoint tool guided by a CNC machine. There are two subtypes of CISF:
 - **Single-Point Incremental Forming (SPIF):** A single tool moves over the sheet surface to create the desired shape.
 - **Two-Point Incremental Forming (TPIF):** Two tools, one deforming and one supporting, work together to shape the sheet.
- 2. Hybrid Incremental Sheet Forming (HISF):** HISF is a modified version of CISF where the sheet is supported by pressurized hydraulic fluid, a partial die, or a full die during the forming process.

Key Process Parameters

The performance of ISF is influenced by various process parameters, including:

- **Tool Path:** The trajectory of the tool significantly impacts the final part geometry and thickness distribution.
- **Tool Geometry:** The shape of the tool (hemispherical, ballpoint, etc.) affects the deformation process and the resulting part quality.
- **Material Properties:** The type and thickness of the sheet metal influence the formability and final characteristics of the part.
- **Incremental Step-Down Size (Step Size, Δz):** This parameter affects both the machine time required for forming and the surface quality of the finished part.

- **Feed Rate:** The speed at which the forming tool moves across the sheet impacts the forming process and material behavior.
- **Forming Angle (ϕ):** The angle between the undeformed and deformed sheet metal serves as a measure of material formability.

2.4. Review of SPIF Research

This section summarizes major research findings on SPIF, including formability limits of various materials, optimal parameter combinations, comparisons to other methods, and the challenges and limitations of SPIF.

Xuelel Zhao and Hengan Ou(12) Proposed a new flexible multi-point incremental sheet forming process with multi-layer sheets (F-MPIF-MLS). This process aims to overcome the problems of excessive sheet thinning and poor surface finish in conventional forming processes. The researchers conducted a comparative investigation between the new F-MPIF-MLS process and conventional MPIF processes through experimental tests and simulations. The results showed a significant reduction in wrinkling using the new flexible multi-point die compared to the conventional multi-point die system with multi-layer sheets to form a dome shape. In addition, the maximum thickness reduction was significantly reduced from 35% to 5% when using the new F-MPIF-MLS process compared to the conventional MPIF process, achieving a more consistent thickness distribution of the formed part.

Cerro et al(13) Investigated a new approach to multi-stage incremental forming, called the multi-stage, two-point incremental forming-rolling blank holder method (TPIF-RL).

The aim of this study was to produce parts with vertical edges and examine the resulting wall thickness distribution.

The experiments were conducted using DC04 sheet metal with a thickness of 0.98 mm, forming an axially symmetrical workpiece with a 90° wall angle. In addition,

The researchers studied the effects of four parameters on the forming process: increment, feed rate, clamping pressure, and angle increment.

The results showed that the thinnest areas of the formed parts were at the lower radius (measurement point 10 in Figure 2.1) and the total thinning amount was a maximum of 0.75 mm.

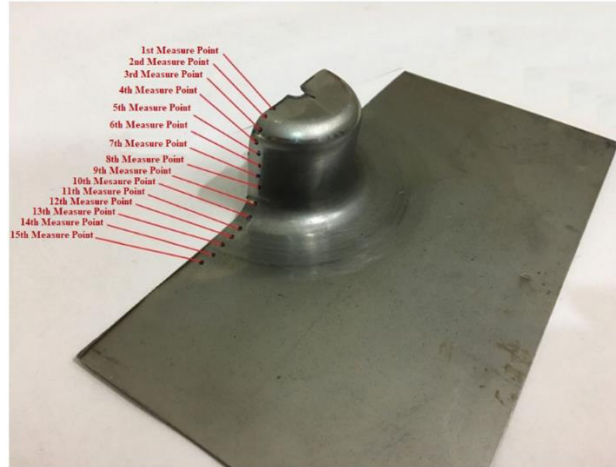


Fig. 2.1 Branded experiment part

The study identified the parameter levels that resulted in the best wall thickness, which were a pressure of 5.5 bar, an angle increment of 10° , a feed rate of 3500 mm/min, and an increment of 0.25 mm.

The research concluded that the multi-stage TPIF-RL method can successfully form parts with vertical edges while minimizing thinning and maintaining the desired wall thickness.

Cerro et al(14) Investigated the effect of evolving anisotropy on the prediction of deformation and forming load in incremental sheet forming (ISF) processes using finite element (FE) modeling. They applied an evolutionary anisotropic plasticity model (e-NAFR Hill's 48), based on Hill's 48 yield function with the non-associated flow rule (NAFR), to the single point incremental forming (SPIF) process of aluminum alloy 6014-T4.

The researchers found that the evolutionary constitutive model showed an average accuracy of 99.15% in thickness prediction and 94.22% accuracy in forming force, demonstrating the importance of evolving sheet anisotropy in the SPIF process.

Cerro et al(15) Studied the single-point incremental forming (SPIF) process using both coated and uncoated tools, guided by an industrial robot.

The aim of this study was to analyze the variation in deformation forces for a pyramid trunk-shaped part, perform ANOVA analysis on the maximum deformation force, and assess the accuracy and surface roughness of the formed parts. The study found that the TiN-coated tool resulted in lower forming forces compared to the uncoated tool, as shown in Table 2.1 and Table 2.2, which compare the maximum forces for both cases.

Case	Feed Rate [mm/min]	Forming Step [mm]	Maximum Force Fz [N]	Maximum Force Fx [N]	Maximum Force Fy [N]
Experimental	5500	0.56	1390	380	400
Simulation	5500	0.56	1395	410	420
Diference [%]			0.36	7.89	5

Table2.1. Numerical and experimental results for forces—case1.

Case	Feed Rate [mm/min]	Forming Step [mm]	Maximum Force Fz [N]	Maximum Force Fx [N]	Maximum Force, Fy [N]
Experimental	5500	0.56	1380	420	440
Simulation	5500	0.56	1490	440	450
Diference [%]			7.97	4.76	2.27

Table2.2. Numerical and experimental results for forces—case2

The researchers also observed deviations in the shape of the formed parts, particularly in the wall curvature, due to variations in forming force, cold hardening, and material thinning.

the surface roughness of the parts formed with the TiN-coated tool was found to be better than those formed with the uncoated tool, as illustrated in Figure 2.2 and Figure 2.3, which show the roughness values for both cases.

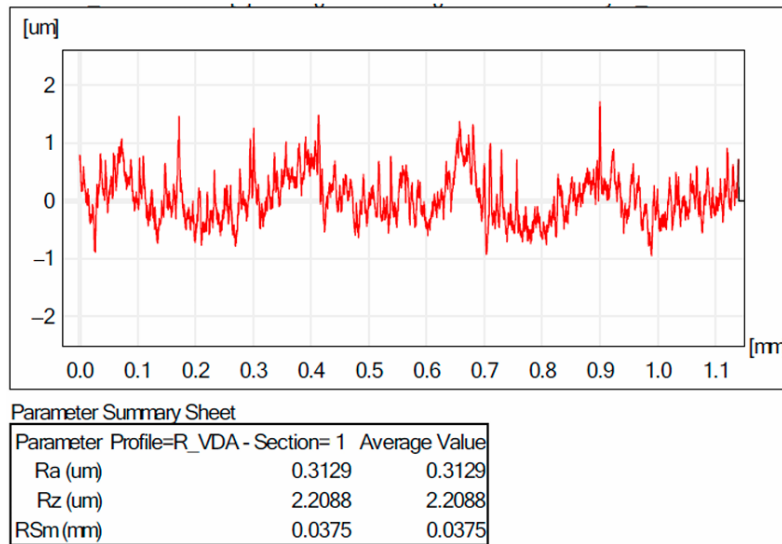


Figure 2.2. Roughness values for the part in experiment 2.

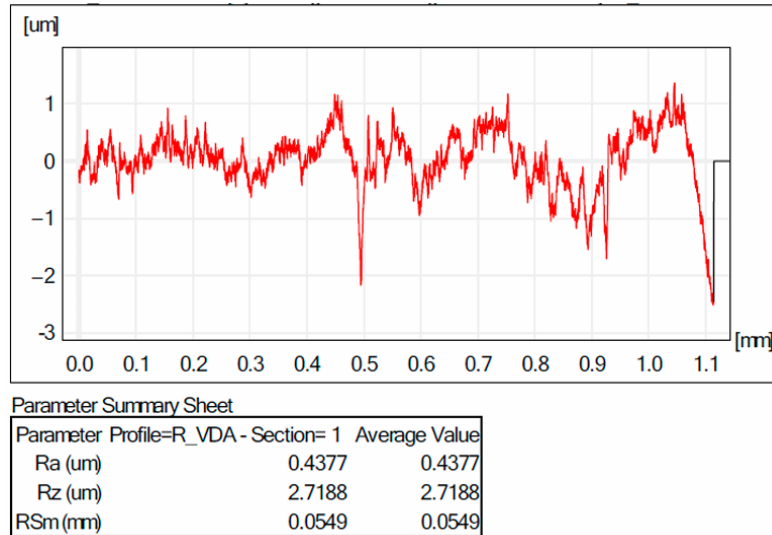


Figure 2.3. Roughness values for the part in experiment 6.

The research concluded that the use of TiN-coated tools in SPIF can lead to lower forming forces and improved surface quality compared to uncoated tools

Cerro et al(16).Studied the effect of tool path strategy on twist behavior in the single point incremental sheet forming (SPIF) process. They developed an adaptive tool path strategy and tested it on several aluminum conical models.

The researchers found that the adaptive tool path strategy had no twisting effect on the final product, while the twisting effect was evident and measurable in the final product when using both iso-planar and helical tool paths when forming a conical cup with a depth of 80 and 110 mm.

The researchers also observed that the forming time using the adaptive tool path was less than the forming time using the helical tool path by 8.7% when forming a conical Cup with a depth of 50 mm and less than 8.91% when forming a conical cup with a depth of 110 mm.

Cerro et al(17)Designed and built a compact and portable dedicated machine for single-point incremental forming (SPIF).

The aim of this project was to create a machine capable of forming small to medium-sized parts with a workspace of 200 mm x 200 mm and a vertical stroke of 100 mm. the machine was designed to primarily form aluminum sheets with thicknesses ranging from 0.5 to 3 mm.

The researchers focused on optimizing the design of the robotic arms, minimizing torque, and analyzing the kinematics of the machine.They carefully selected and adapted components, including the forming punch support system, motors, reducers, and linear actuators.

The machine's structure was designed using 45x45 mm aluminum profiles for an optimal balance of weight and strength. Simulations were conducted to verify the feasibility and performance of the design, ensuring that the machine could withstand the forces involved in the SPIF process.

The researchers successfully built and tested a functional early prototype of the compact incremental forming machine, as shown in Figure 2.4.

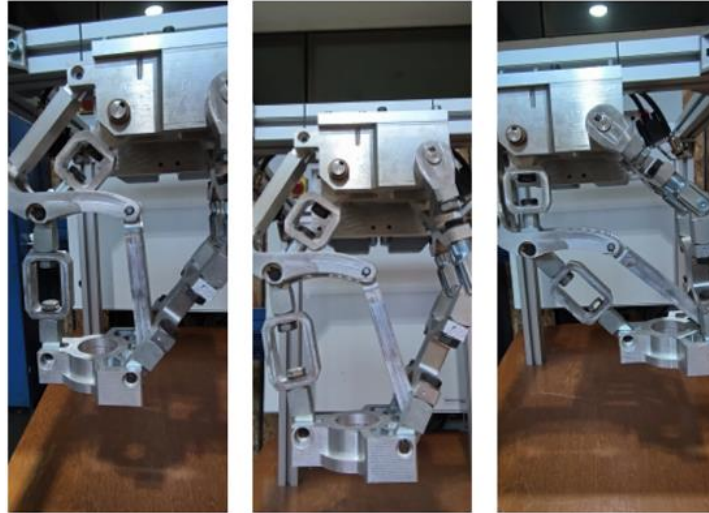


Figure 2.4. Prototype of the kinematic system.

Cerro et al(18) Investigated the effect of process parameters such as tool diameter, step size, and spindle speed on the average surface roughness of formed components on AA2024-O sheets.

Truncated cone-shaped objects were formed to study the influence of tool diameter, step size, and spindle speed. The researchers found that average roughness increases with the decrease in tool diameter, and that the combination of a larger tool diameter and smaller step size produced a better surface finish of the components.

Hui Zhu and Hengan Ou, Developed (19) a new heat-assisted friction stir incremental sheet forming (HAFS-ISF) process to form polyether ether ketone (PEEK) and polymethyl methacrylate (PMMA).

The researchers conducted a comparative investigation between the new HAFS-ISF process and conventional ISF processes through experimental tests and simulations. The results showed a significant improvement in temperature distribution and a reduction in the temperature gradient between the upper and lower surfaces of the sheets.

The results also showed that spindle speed has the most significant effect on temperature, forming force, dimensional accuracy, and thickness distribution, while feed rate and step size have less effect.

Omer Secgin and Ibrahim Ozsert(20) Investigated the formability of Titanium Grade 2 (Ti Gr 2) sheet using a newly developed method called the roller blank holder method (TPIF-RL). The aim

of this study was to optimize the forming process to achieve greater wall thickness with lower clamping force compared to conventional methods.

They conducted experiments using parameters such as feed rate, clamping pressure, increment, and forming tool diameter, and analyzed the resulting forming force, surface roughness, and wall thickness distributions.

The study found that the TPIF-RL method, when using a lower clamping pressure (2 bar), resulted in a greater wall thickness compared to conventional incremental forming methods. The optimal levels for the parameters were determined using the Taguchi method and grey relational analysis, with the goal of optimizing all three dependent variables (forming force, surface roughness, and sheet thickness) simultaneously.

The research concluded that the TPIF-RL method can effectively form Ti Gr 2 sheets with improved wall thickness, which enhances the usability of the final part.

Cerro et al(21) Focused on developing a MATLAB code to generate tool paths and G-codes for various hole-flanging geometries in single-point incremental forming (SPIF). The aim of this study was to provide a unified programming technique within the MATLAB environment to streamline the tool path planning process for incremental hole flanging. The researchers considered six different hole flange geometries, including tapered holes, circular holes, tapered square holes, square holes, and U-shaped hole flanges with and without tapered holes.

The MATLAB code successfully generated 3D tool paths and corresponding G-codes for each geometry, as shown in Figure 2.5. These G-codes can be directly used for CNC machining or integrated with time data for FEM simulations.

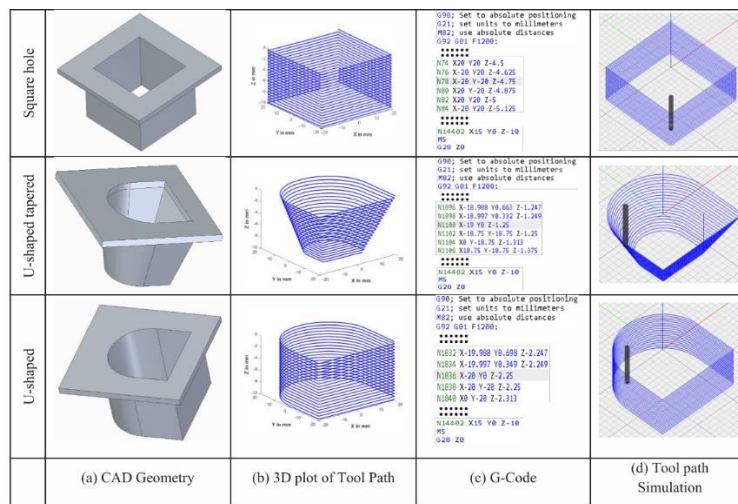


Fig. 2.5. (a) CAD geometry (b) 3D plot of Tool Path (c) G-Code, (d) Tool path Simulation.

The proposed methodology offers a more efficient and time-saving approach compared to existing methods, as it combines tool path generation and G-code creation in a single MATLAB

program. The researchers concluded that this methodology can be effectively applied to various intricate shapes in incremental hole flanging, demonstrating its potential for broader applications in sheet metal forming.

Yogesh Kumar and Santosh Kumar(22)Focused on the incremental sheet forming (ISF) process, a flexible manufacturing process used to create complex 3D products without dedicated tools.

They developed a single-point incremental forming machine using a motion card to control three servo motors for 3-axis movement and spindle movement.

The study investigated the strain distribution on the sheet during the deformation process and assessed the surface quality of the formed products.

The researchers also performed simple simulations to analyze the ISF process further.

The results showed that the strain was distributed normally over the length of deformation. In addition, the simulation findings, and defect-free products were successfully produced using the developed ISF machine.

The research concluded that the ISF process is a viable method for producing complex 3D shapes with good surface quality, and the developed machine can be used effectively for this purpose.

Cerro et al(23)Focused on analyzing the damage evolution and ductile fracture in incremental forming at a microscopic level.

They extended the Gurson-Tvergaard-Needleman (GTN) model by integrating Hill'48 anisotropic yield criterion and implemented it in ABAQUS using VUMAT.

The study involved a single element test for verification and a tensile test to determine damage parameters. A finite element model of incremental forming for a truncated cone was established and experimentally verified.

The influence of two damage parameters, f_N (volume fraction of void nucleating particles) and ϵ_N (mean plastic strain value when voids nucleate), on damage evolution was explored.

The results showed no significant connection between microscopic equivalent strain (ϵ_{plm}) and f_N or ϵ_N values. However, the maximum void volume fraction increased with higher f_N or lower ϵ_N values.

The void volume fraction was proven to be an effective index for predicting crack occurrence in incremental forming, validated by fracture tests on truncated cones with different wall angles (Figure 2.6).

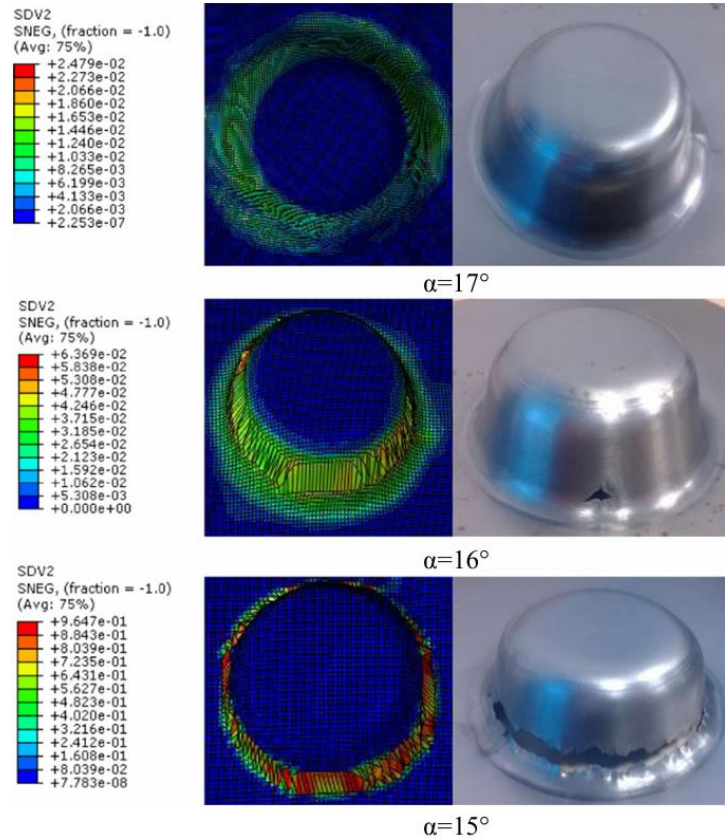


Fig. 2.6. Simulation results using Hill'48–GTN model and experimental results at different wall angles

Cerro et al(24) Focused on heat-assisted single-point incremental forming (SPIF) for high-strength materials. The aim of this study was to review the heating-assisted SPIF systems for high-strength alloy sheets to solve the current limitations.

They discussed various heating methods used in SPIF, including friction stir, ultrasonic vibration, electric heating, laser heating, and induction heating. Each heating method has its advantages and limitations. For example, friction stir is a low-cost method but is not efficient for high-strength materials, while laser heating offers precise temperature control but is expensive.

The research also explored tool design, highlighting the benefits of ball-roller tools over conventional tools in reducing friction and improving formability. This is illustrated in Figure 2.7, which compares the strain distribution between the ball-roller tool and workpiece.

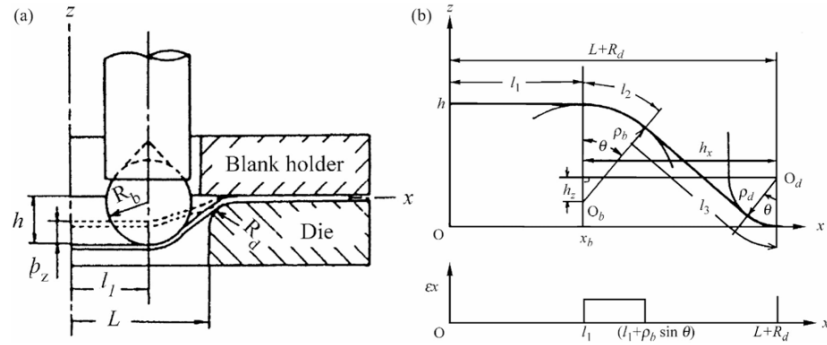


Fig. 2.7 Ball-roller tool and working mechanism. a Section view of the ball-roller tool, b plane-strain deformation of the ball-roller tool

Tool path design is another crucial aspect discussed in the paper, with contour and helical tool paths being compared. Additionally, the study delves into tool path optimization techniques like compensation, RSM, ANN, and image analysis. Lubricants and coatings are also examined, emphasizing the need for specialized lubricants in heat-assisted SPIF due to the rapid dissipation of conventional lubricants at high temperatures.

The paper concludes by summarizing the findings and suggesting future research directions, such as developing localized and stable heating methods, designing functional tools to minimize thermal expansion and friction, and analyzing the relationship between thermal and mechanical effects in heat-assisted SPIF.

Chapter 3

EXPERIMENTAL WORK

3.1 Overview

This chapter details the experimental work, including the properties and chemical composition of the workpiece material, the fixing method used. specifications of the forming tool, forming conditions, and the experimental procedure. Additionally, it provides an in-depth discussion of the measurement procedure and measurement devices used for evaluating surface roughness and sheet thickness

3.2 workpiece material

The material used in the present study is aluminum alloy **AA1050**. The aluminum alloy was supplied in sheet form with a thickness of 0.9 mm and precisely cut to the specified dimensions of 200 x 200 mm.

3.3 setup design

The primary advantage of CNC forming lies in the simplicity of its tooling, which grants the process exceptional flexibility. Unlike traditional forming techniques that necessitate intricate and time-consuming die fabrication, CNC forming operates without the need for dies. Fixture design is influenced by three key factors: the deformed sheet, the machine type and its limitations, and the complexity of the product.

For CNC forming, a specialized fixture was employed to secure the sheet metal during the forming process. This fixture was designed and manufactured with appropriate dimensions for this specific application, as illustrated in Fig. 3.1. The design of the fixture took into account several parameters, including the shape and size of the deformed sheet, the machine's operational limits, and the complexity of the part to be produced.

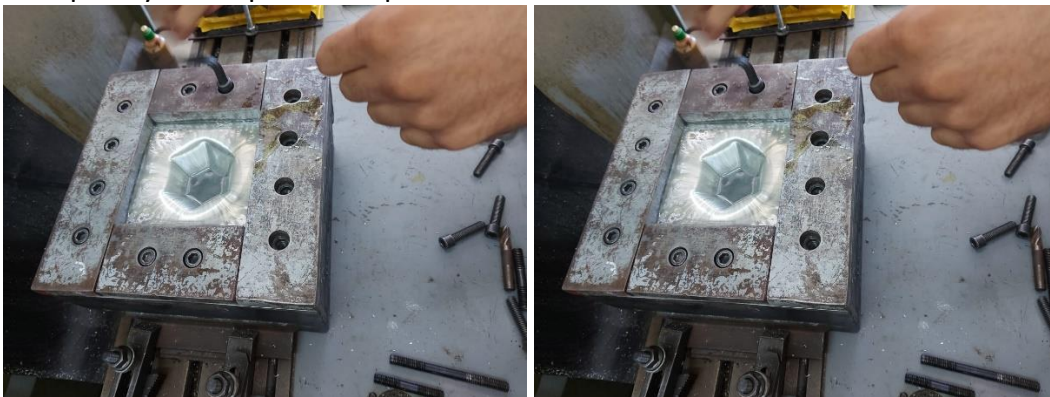


Figure 3.1 the fixture components

3.4 forming tool

The forming tool is the component that transforms the CNC milling machine's action into the sheet metal forming process. In this study, a tool steel bar (K100) with a high carbon content and 12% chromium, measuring 250 mm in length and 15 mm in diameter, was utilized. This bar was machined down to a length of 100 mm and its end was turned into a hemispherical shape with a 12 mm diameter, as depicted in Fig. 3.3. The choice of K100 tool steel ensures durability and resistance to wear, essential for maintaining precision and efficiency in the forming process.

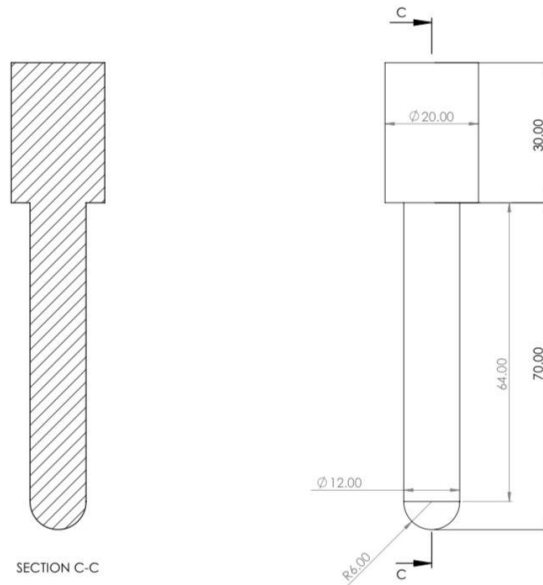


Figure 3.2 Forming Tool

3.5 sample deformation

Hexagonal shape was chosen for the experimental work and finite element modeling (FEM). The forming process was conducted using an Extron CNC milling machine, as shown in Fig. 3.4a. Fig. 3.4b illustrates the forming process used to create the hexagonal shape of the final part.



Figure 3.3 Extron CNC milling machine and set up



Figure 3.4 CNC deformed shape

3.6 forming parameters

Single point incremental forming process (SPIF) was performed the experimental investigation to study the process behavior and the effect of process parameters on the process outcomes. In particular, four different feed rates in y-direction were selected to perform the experiments, the feed rate varied between 1000 and 4000 mm/rev with a step of 1000 mm/min, and the vertical tool increment was ranged between 0.2 and 0.8 mm with a step of 0.2 mm.

All specimens were carried out at two different spindle speeds of 800 and 3200 rpm for all feed rates, and incremental depths. However, Each deformed specimen was previously coded by (α , z, n), where α is a constant shear angle of 57.5°, z is the incremental depth, and n represent the tool spin rpm.

Constant parameter material
(A.A.1050), Sheet thickness=0.8

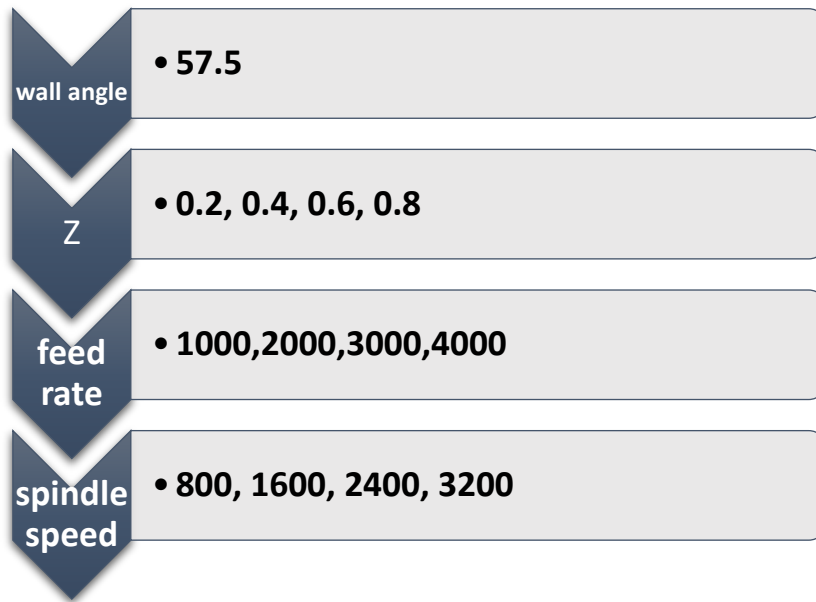


Fig. 3.5: Experimental parameters flow chart

3.7 Forming processing

The main principle of the CNC forming process is to obtain the desired shape of the product by the relative movement of a simple tool in relation to the sheet blank, without using dies. A square metal blank of 200 mm side is clamped around its contour in order to restrain any movement during forming process. A hemispherical ended steel tool used for stretch forming is mounted on the spindle. In this process, a layer at constant depth in axial direction is formed by an in-plane movement of the tool. After performing a layer, the tool moves down with a small increment along the axis, to process the next layers till the completeness of the process. Fig.3.6 illustrates the main setup parts in CNC forming process.

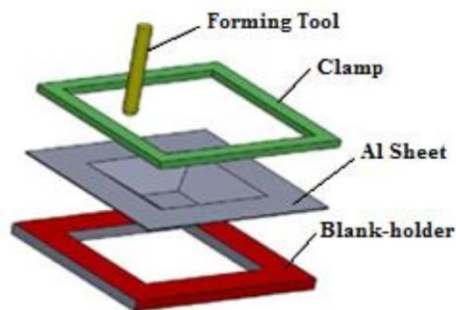


Fig. 3. 6: Forming parts

In the beginning of the experimental work a CAM file (.txt). This file is uploaded to 3D CNC milling machine to deform a part. Extra High Performance Diesel Engine Oil (Mobil Delvac 1300 Super 15W-40) was used for lubricant; large amount of lubricant where the blank was merged, consequently reduce the friction heat and the tool wear as shown in Fig. 3.7.



3.7 Diesel oil used as a Lubricant

3.7 tension test

In the way to get the required material properties of the used material, uniaxial tensile test has been executed in computerized universal testing machine. The tensile test specimen and procedures of Al 6061 of 1.5 mm thickness is tested according to ASTM E8M codes and then experimentally used in the research as sheet specimen

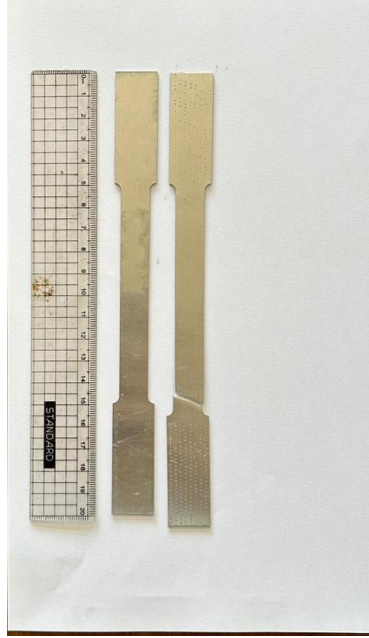


Figure 3.8 Tensile test specimen before and after

Tensile testing is fundamental to material characterization, providing critical insights into mechanical properties such as tensile strength, yield strength, elongation, and Young's modulus

Material	AA1050
Yield strength	25 MPa
Tensile strength	75 MPa
Elastic modulus	68.9 GPa
Poisson's ratio	0.33
Mass density	2.71 g/cm ³ (or 2710 kg/m ³)
Shear modulus	26 GPa
Thermal expansion Coefficient	23.6×10 ⁻⁶ per °C

3.8 FEM Analysis

In the last decades FE analysis was consuming large times and efforts but after huge work, there are some commercial software, such as LS-DYNA, ASNYS, DYNAFORM, DYNA-DIE ABAQUS are available for modeling and simulating these processes. A lot of fundamental researches on metal forming are conducted by FEM simulation for investigating the deformation mechanics and material behavior.

In order to convert the physical problem into finite element model, it is required to idealize the problem accurately where the model must be suitably defined to problem behavior. Choosing forming conditions which were used in previously measured data is the first step. The second step is to generate the tool path of the simulated product. As FEA is a virtual simulation of process it also requires to be related to

the physical core, flow curve and material properties is a conjunction between the actual deformed material and the target model, Fig. 3.8 shows the flow curve of material used AA1050.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Overview

This chapter presents the results of experimental work carried out within this research study. In particular, the effect of forming process parameters on the forming performance is discussed that has been assessed by quantifying the obtainable surface roughness (R_a) and reduction of thickness. The chapter extends detailed discussions of the inputs/outputs relationship and pays more attention to explain the rationale behind the main findings. Besides, the chapter presents the experimental validation of the model. In addition, the chapter introduces the results of the optimum network structures, which established a relationship between forming parameters and forming performance, based on experimental results. Also, the results of Genetic algorithm optimization for the process are discussed. Finally, the FEM results are introduced.

4.2 Effect of incremental sheet metal forming parameters on surface roughness and sheet thickness.

The variations in roughness and thickness values of the deformed surface depending on ISMF parameters such as spindle speed (rpm), feed rate (mm/min), and incremental depth (mm)

We used 9 specimen in our run and this table shown value of each parameters

s	Spindle speed (rpm)	ed rate (mm/m)	step size (mm)
1	1600	2000	0.4
2	1600	3000	0.6
3	1600	4000	0.8
4	2400	2000	0.6
5	2400	3000	0.8
6	2400	4000	0.4
7	3200	2000	0.8
8	3200	3000	0.4
9	3200	4000	0.6

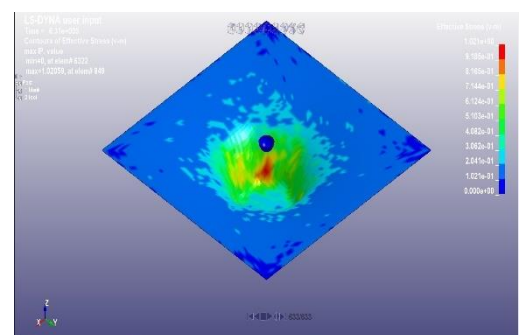
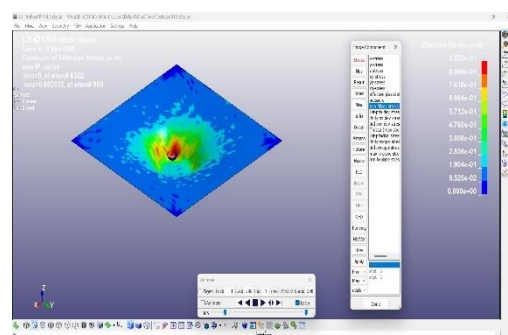
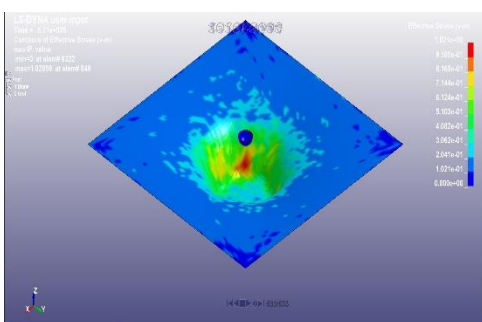
4.2 Effect of stress on sheet metal for 9 specimens (von mises):

Von Mises is a term used in material mechanics, referring to the "Von Mises yield criterion." This criterion is based on the idea that material failure occurs when the equivalent stress (Von Mises stress) reaches a certain limit, which is equivalent to the yield stress in a uniaxial tensile test. The equivalent stress is calculated from the three principal stress components of the material and is useful for assessing the safety and design of engineering structures under multidimensional loads. The Von Mises criterion is commonly used in finite element analysis (FEA) and in civil and mechanical engineering to evaluate potential material failure

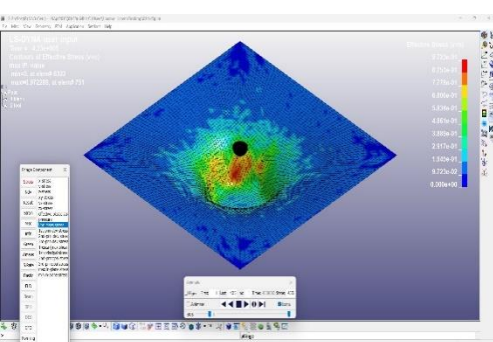
(1)

(2)

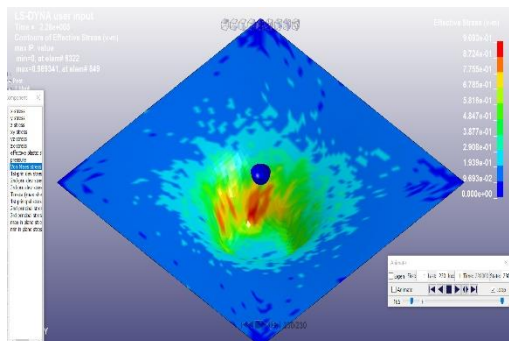
(3)



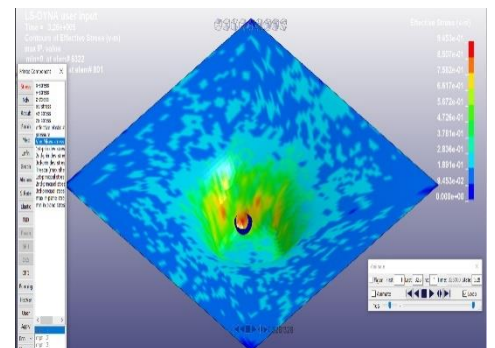
(6)



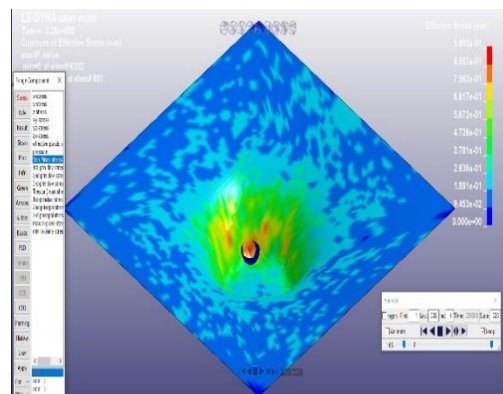
(5)



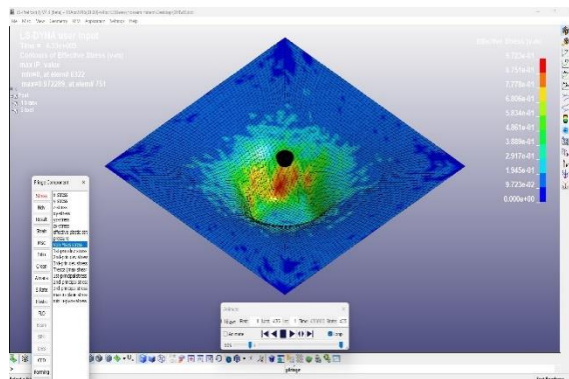
(4)



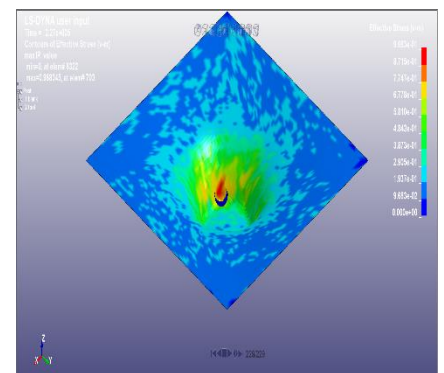
(9)



(8)



(7)



4.3 effect of plastics strain on sheet metals for 9 specimens :

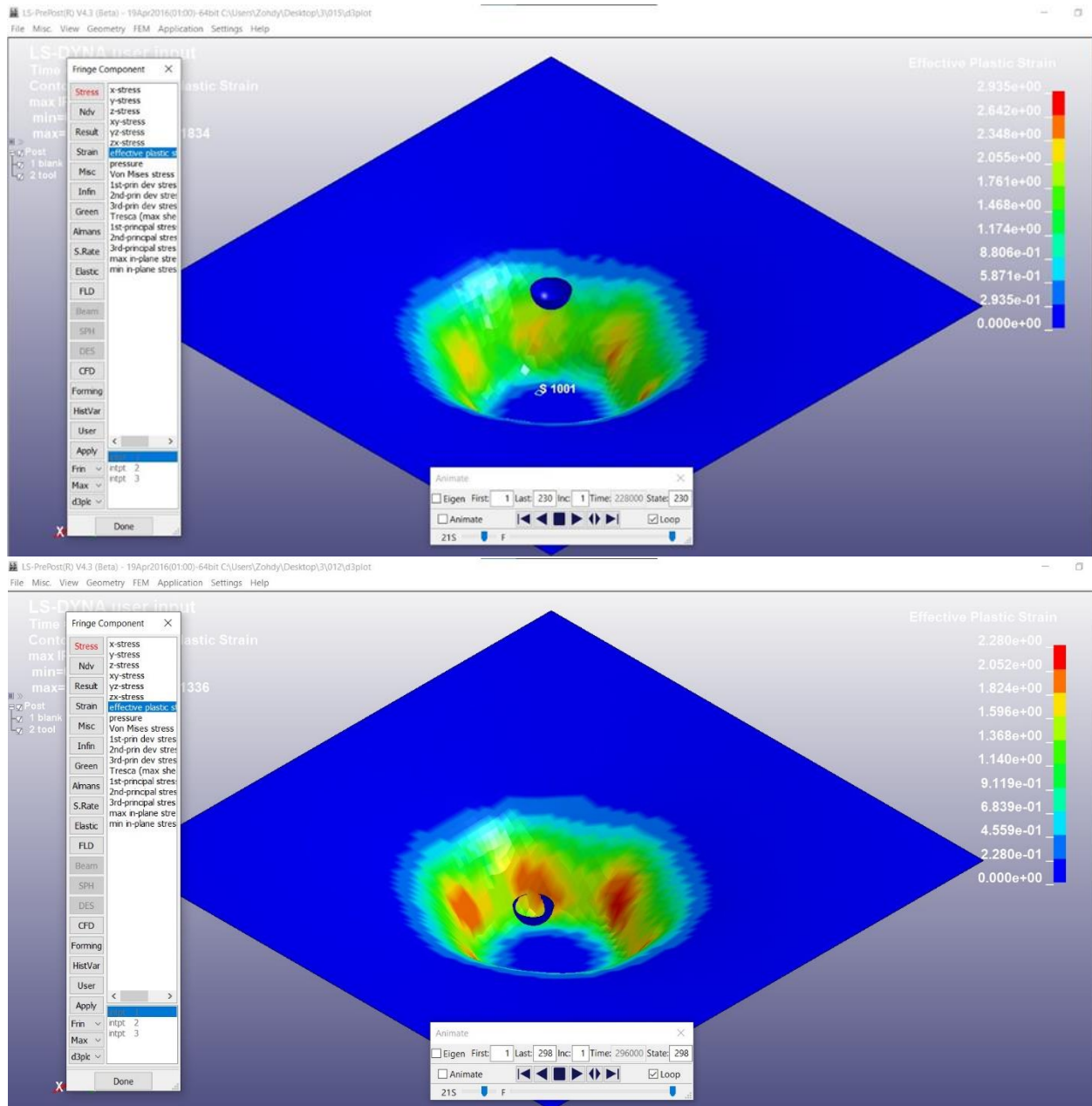
Plastic strain is the permanent deformation of a material under the influence of an applied force after the force is removed.

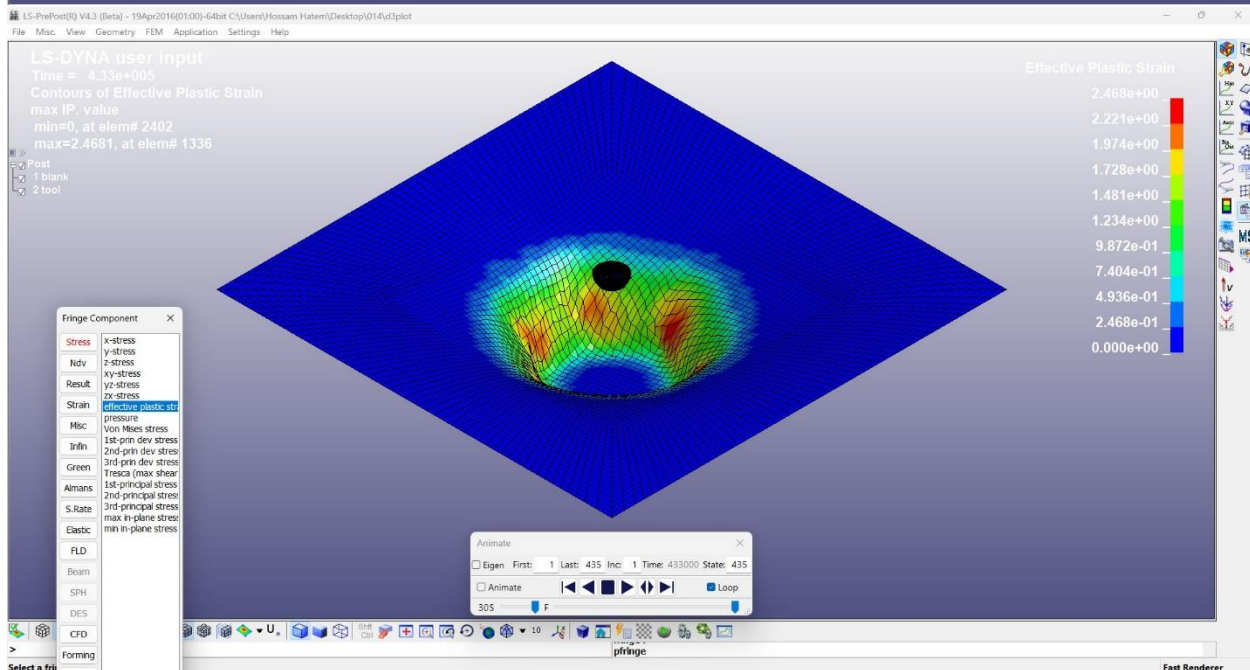
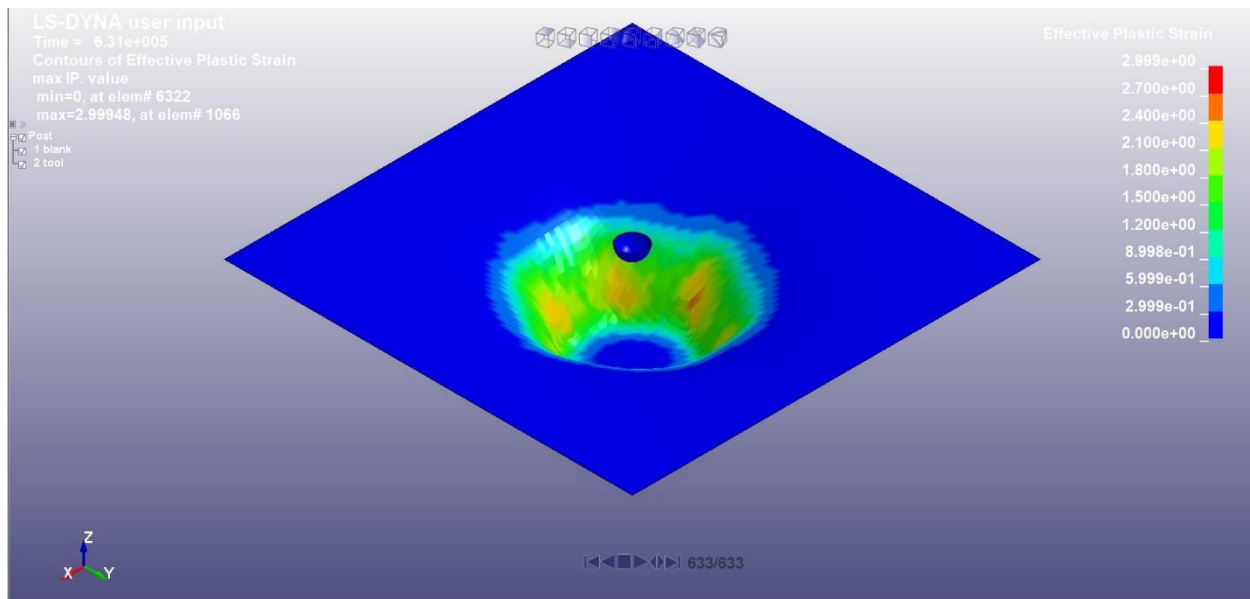
It occurs when the material exceeds its elastic limit, meaning it does not return to its original shape once the load is removed.

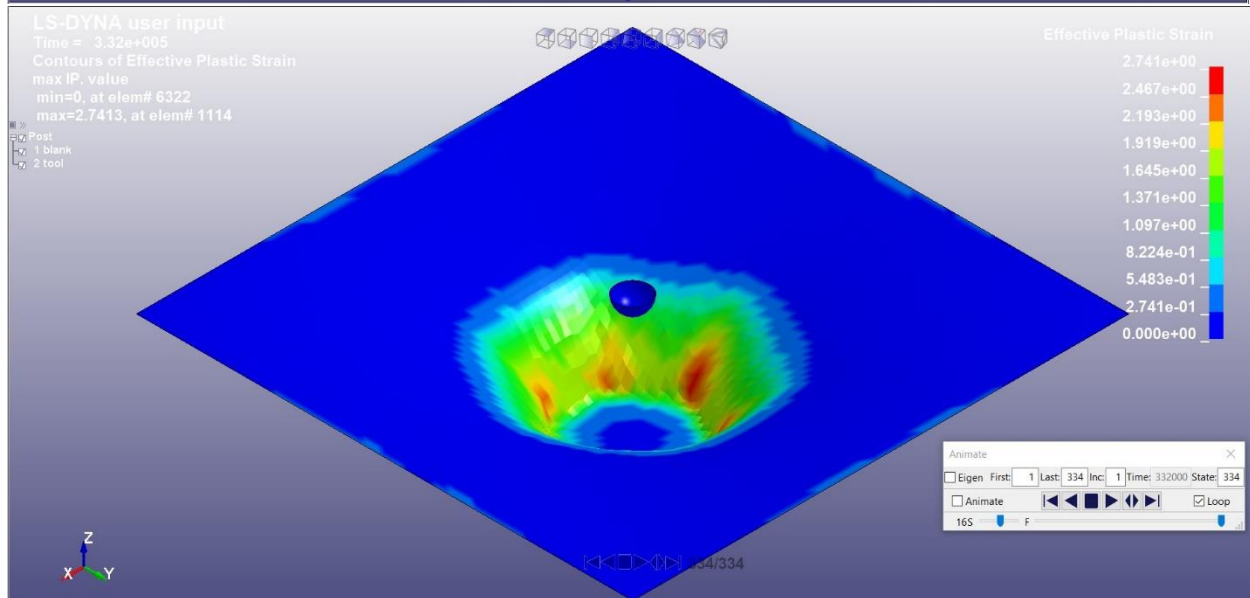
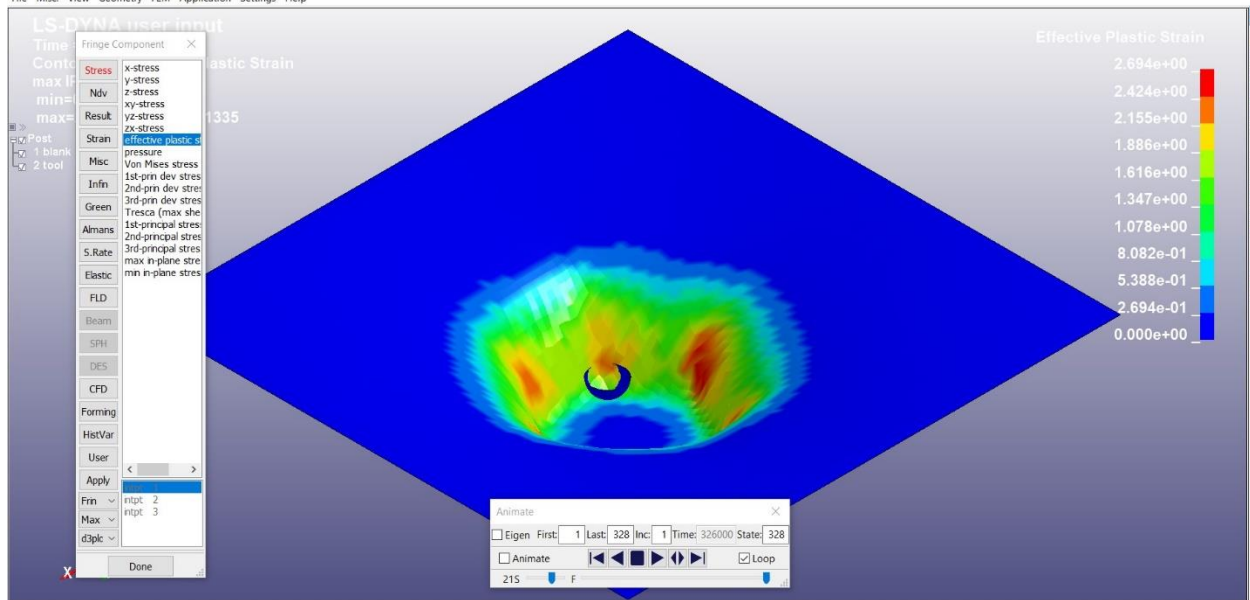
This type of strain is important in the design of engineering and structural materials as it determines the material's ability to withstand deformation without failure.

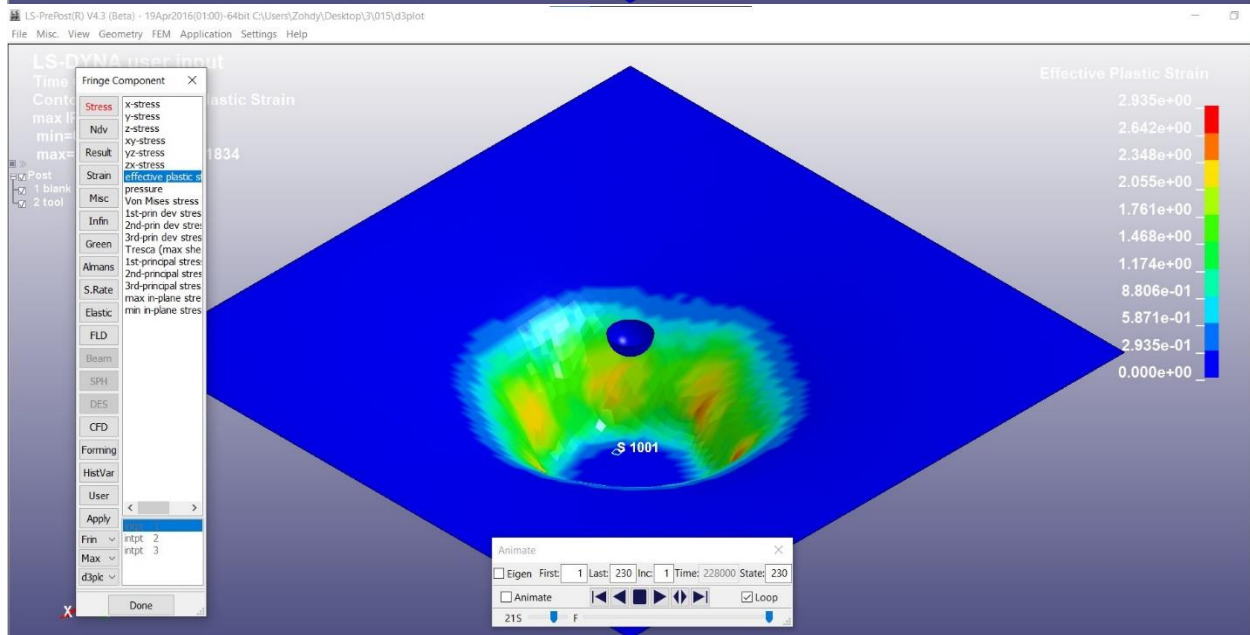
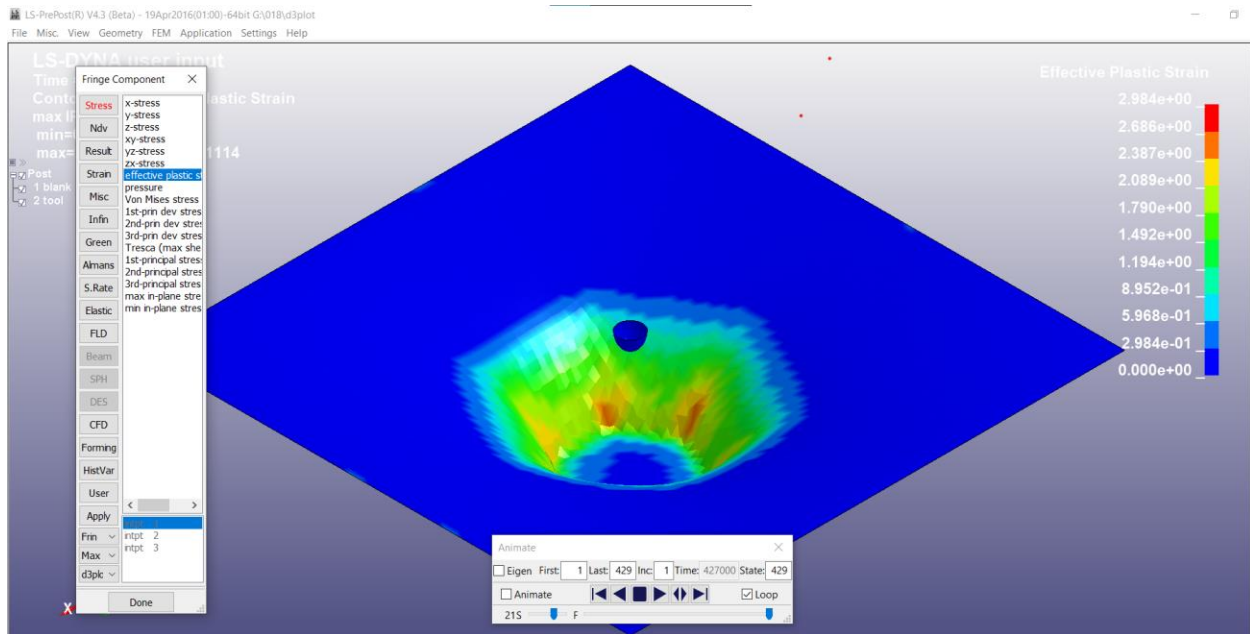
Plastic strain is measured using tensile and compression tests on material samples.

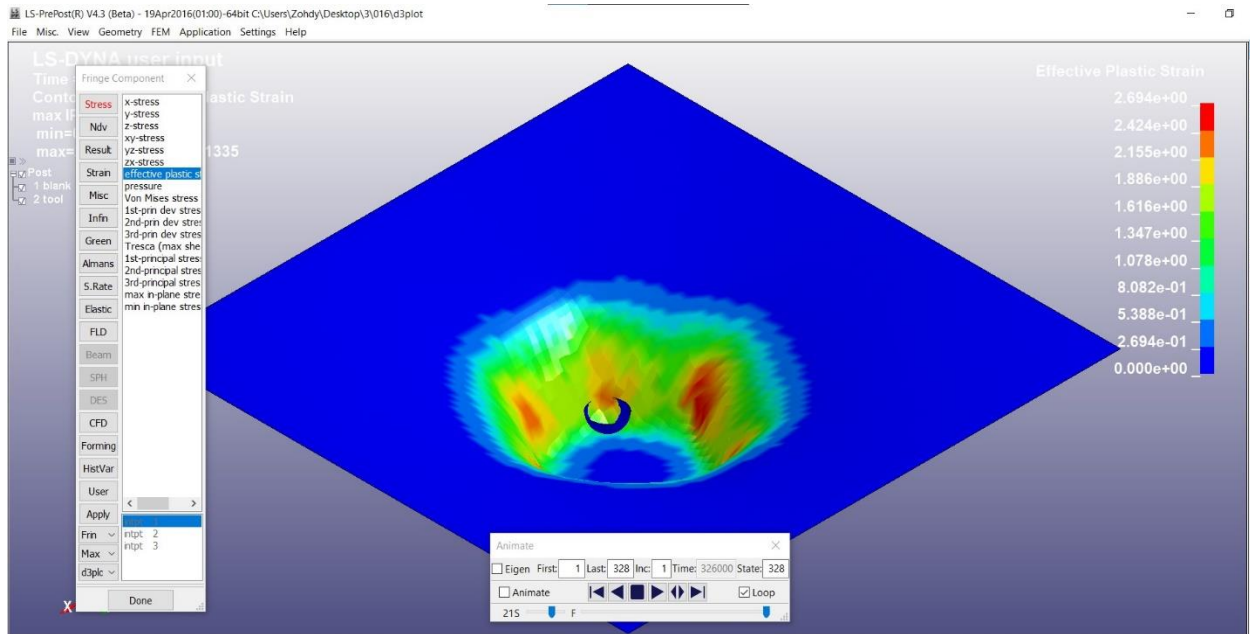
Understanding plastic strain helps engineers improve the performance of materials in various applications.







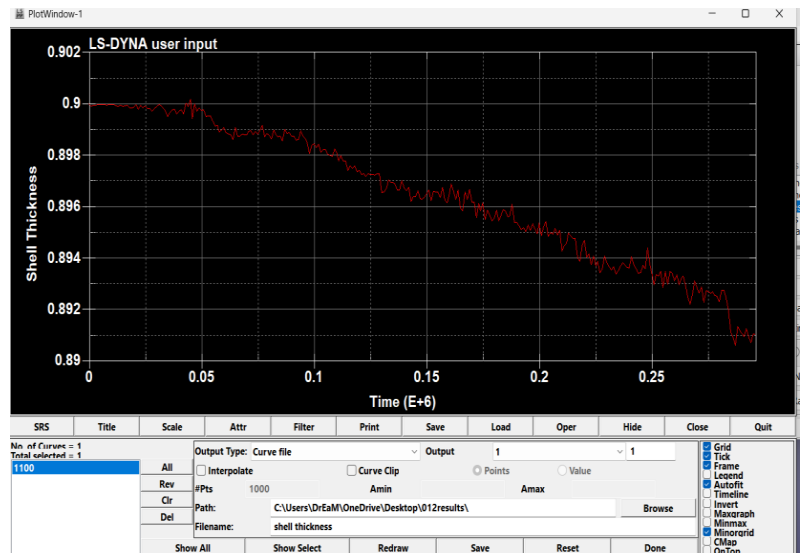
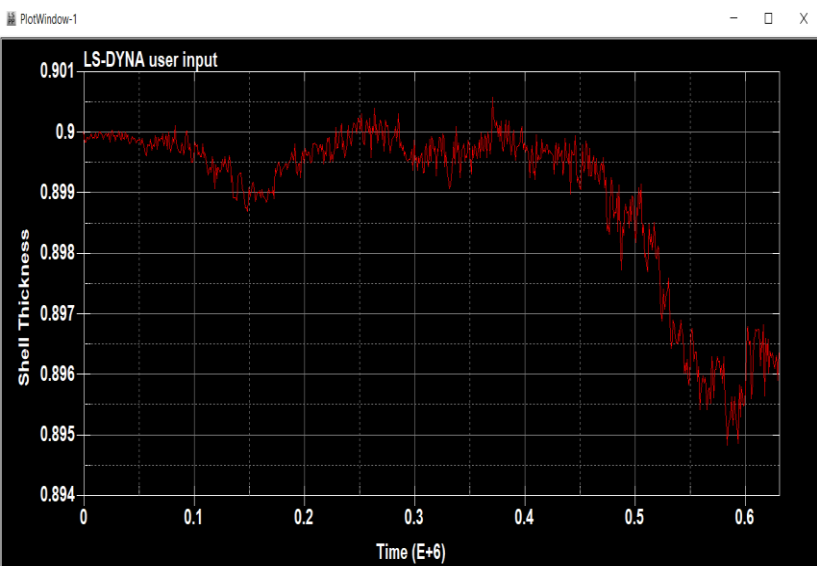


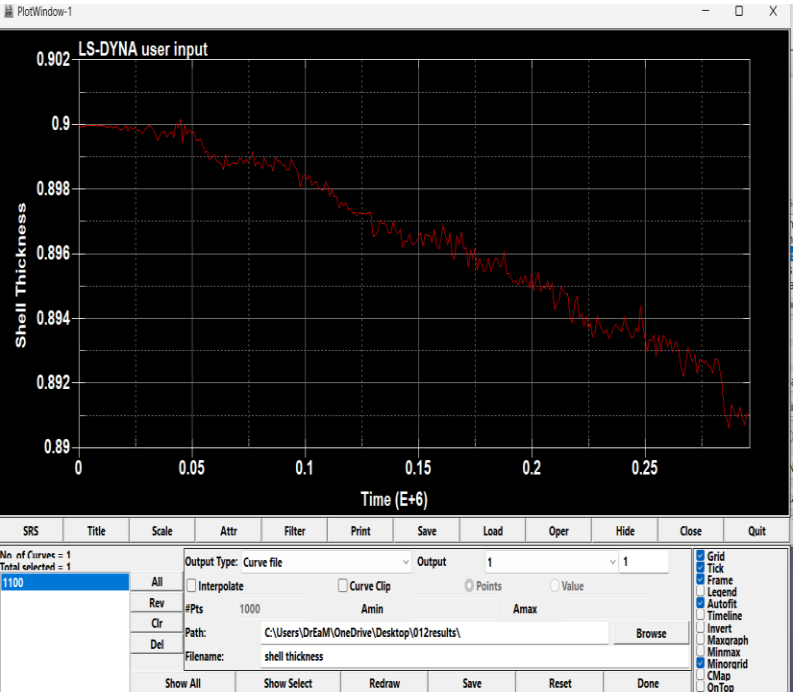


4.3 effect of shell thickness on sheet metals for 9 specimens:

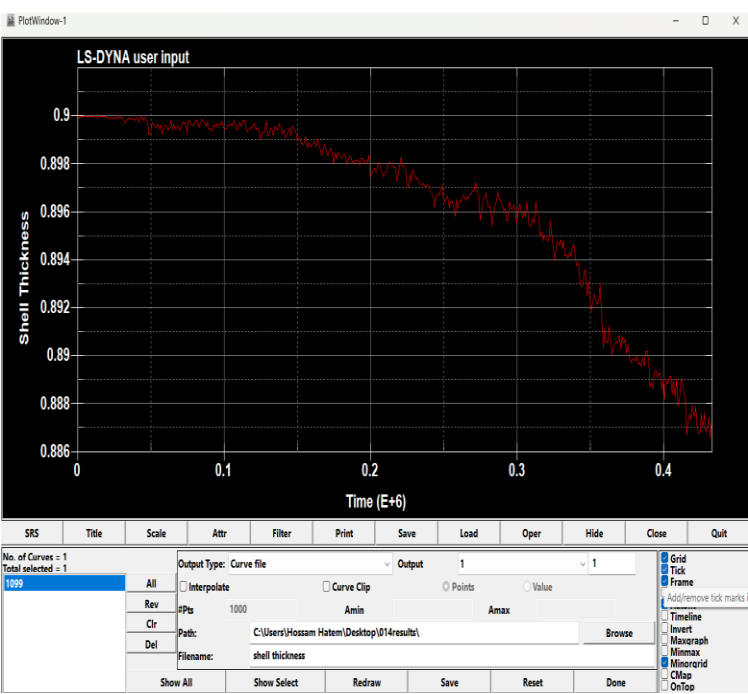
(1)

(2)

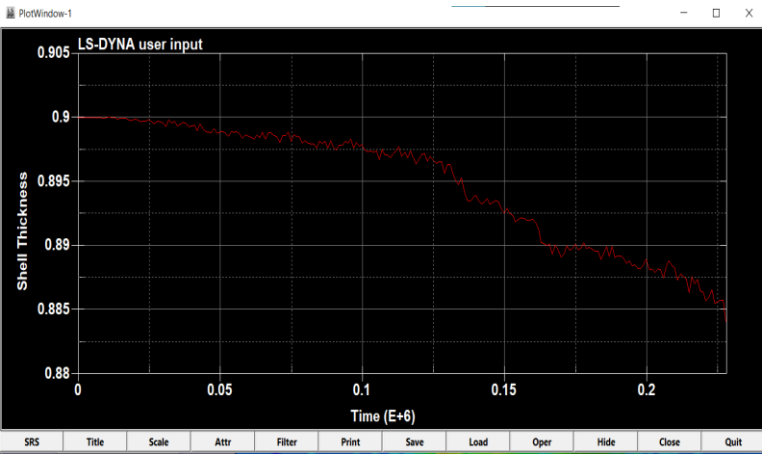




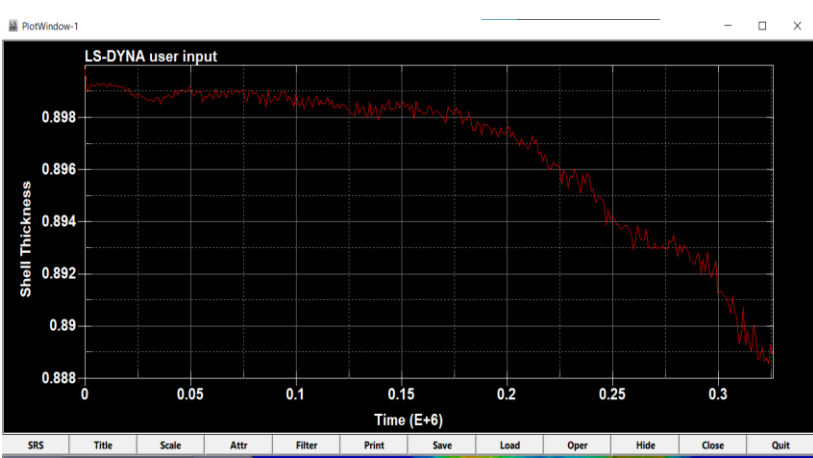
(5)



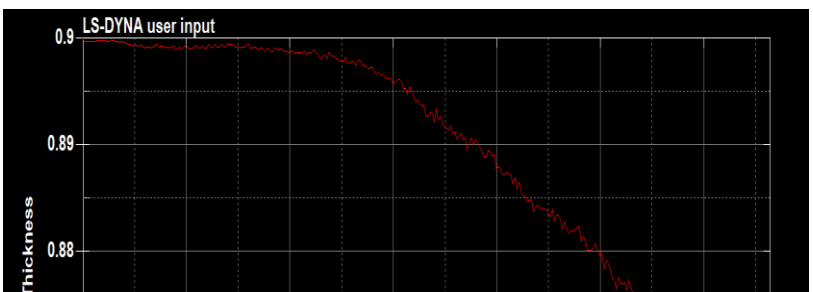
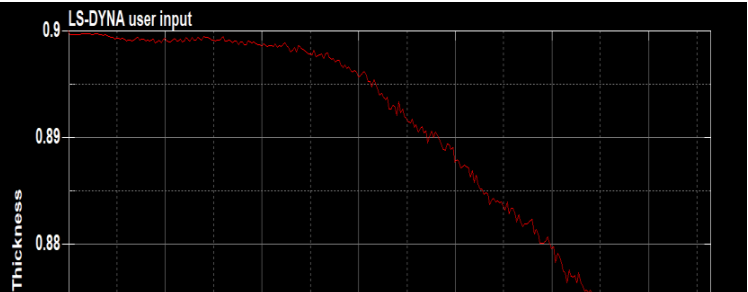
(6)

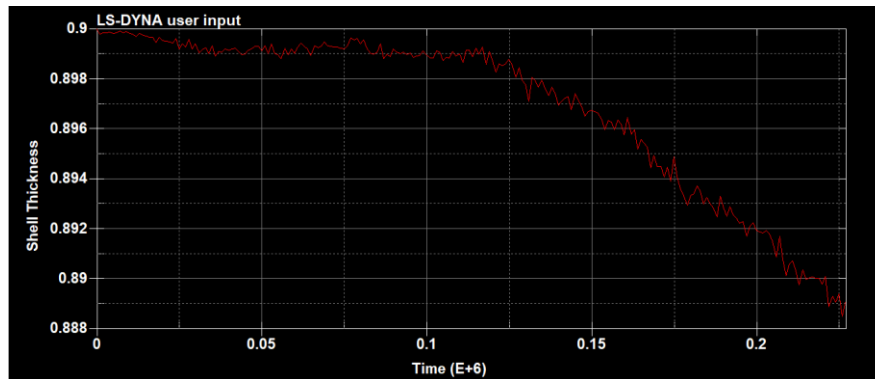


(8)



(7)





4.3 effect of displacement on sheet metals for 9 specimens:

X and z displacement

Displacement is the change in the position of an object from one point to another.

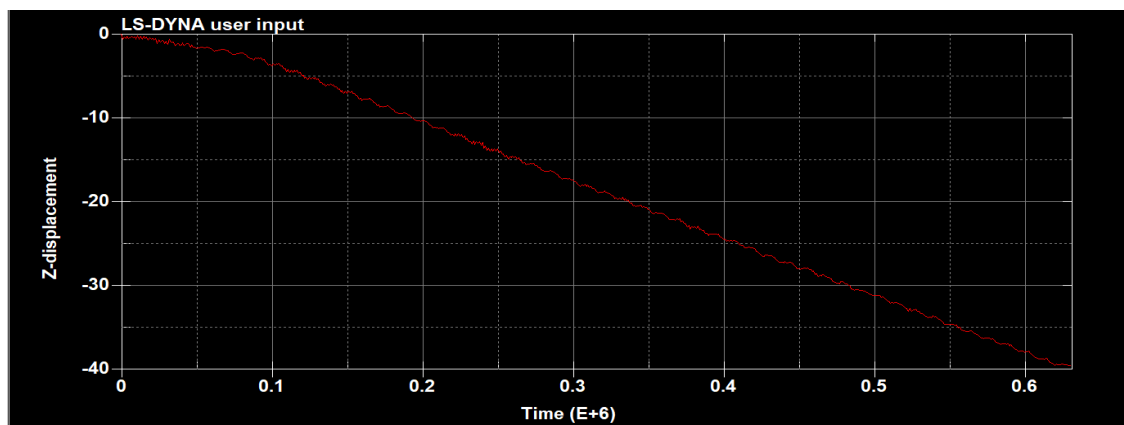
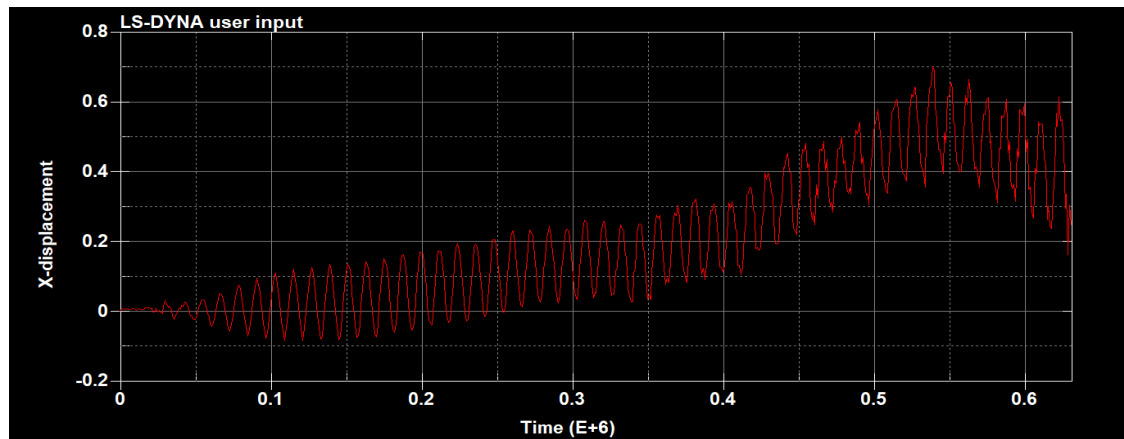
X displacement (horizontal displacement) refers to the amount of change in the object's position in the horizontal direction (left or right).

Z displacement (vertical displacement) refers to the amount of change in the object's position in the vertical direction (up or down).

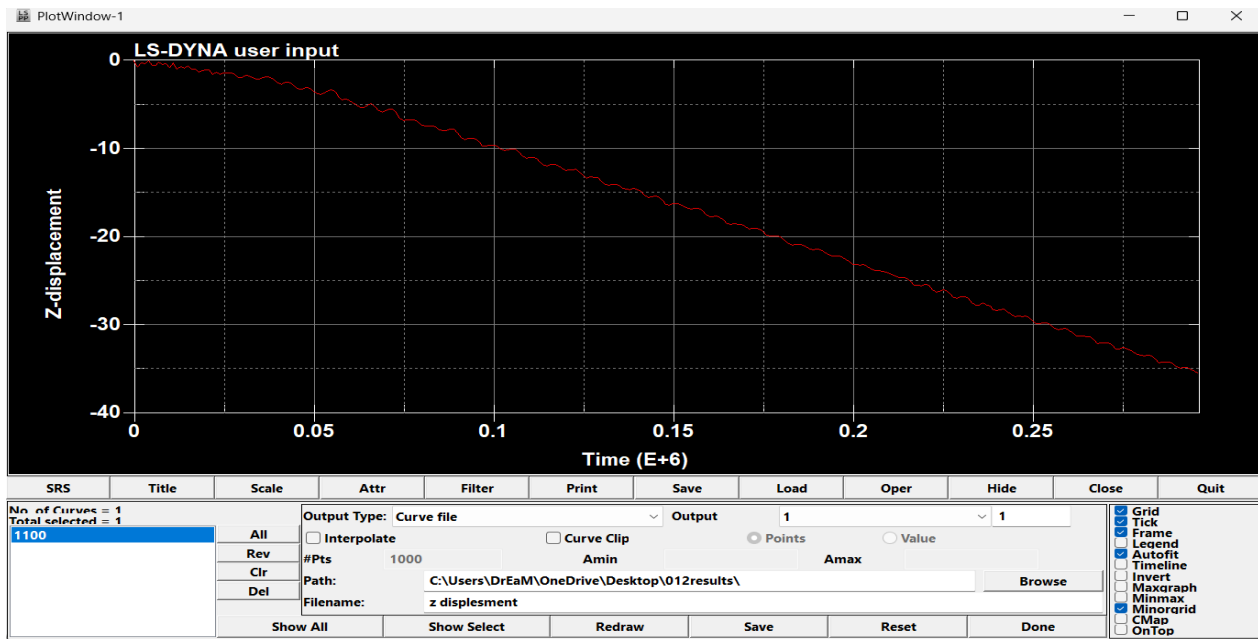
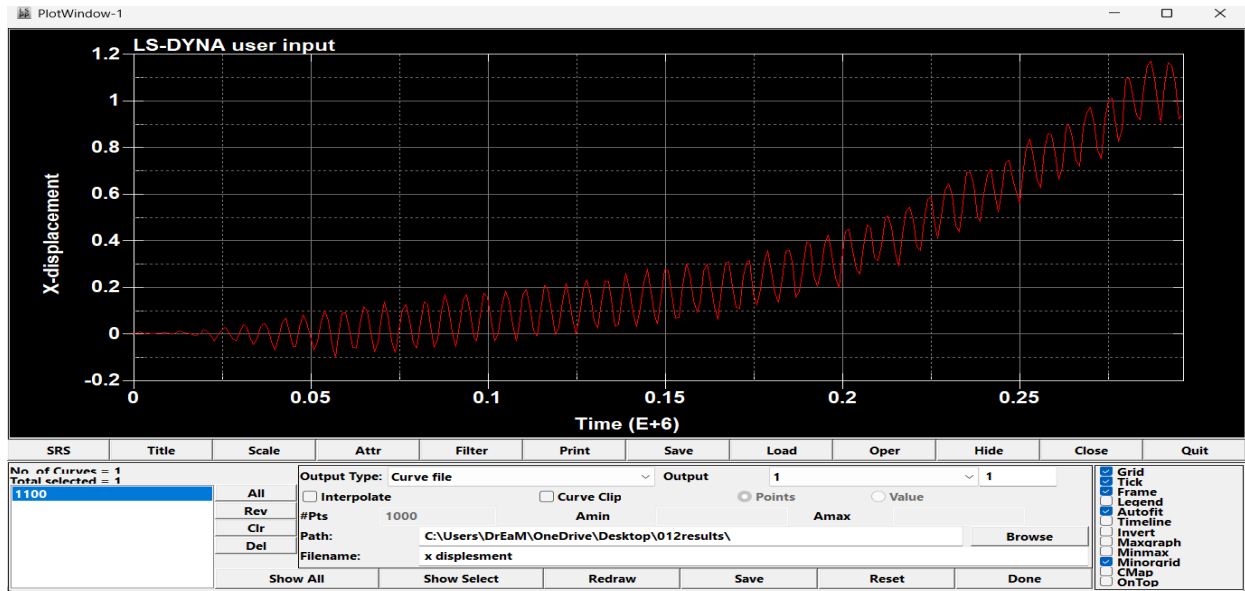
The total displacement of an object moving in three dimensions can be calculated using the Pythagorean Theorem.

Displacement is a vector quantity, meaning it takes into account both magnitude and direction.

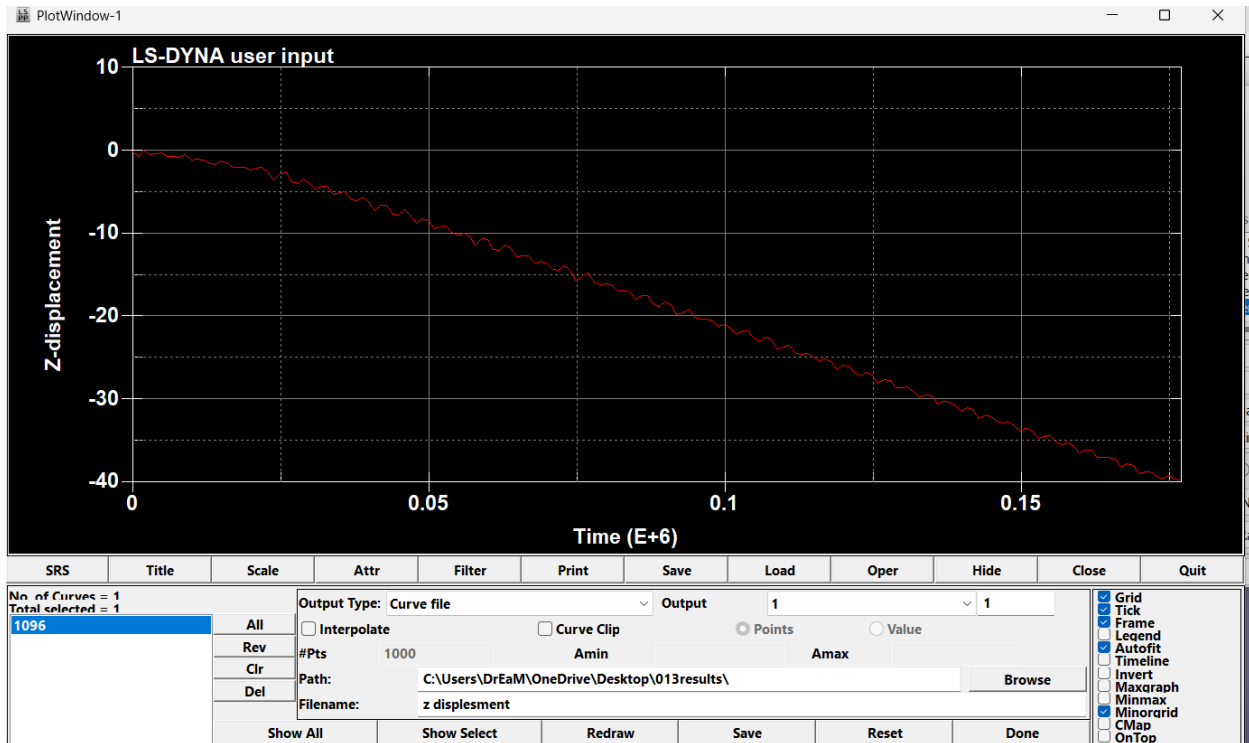
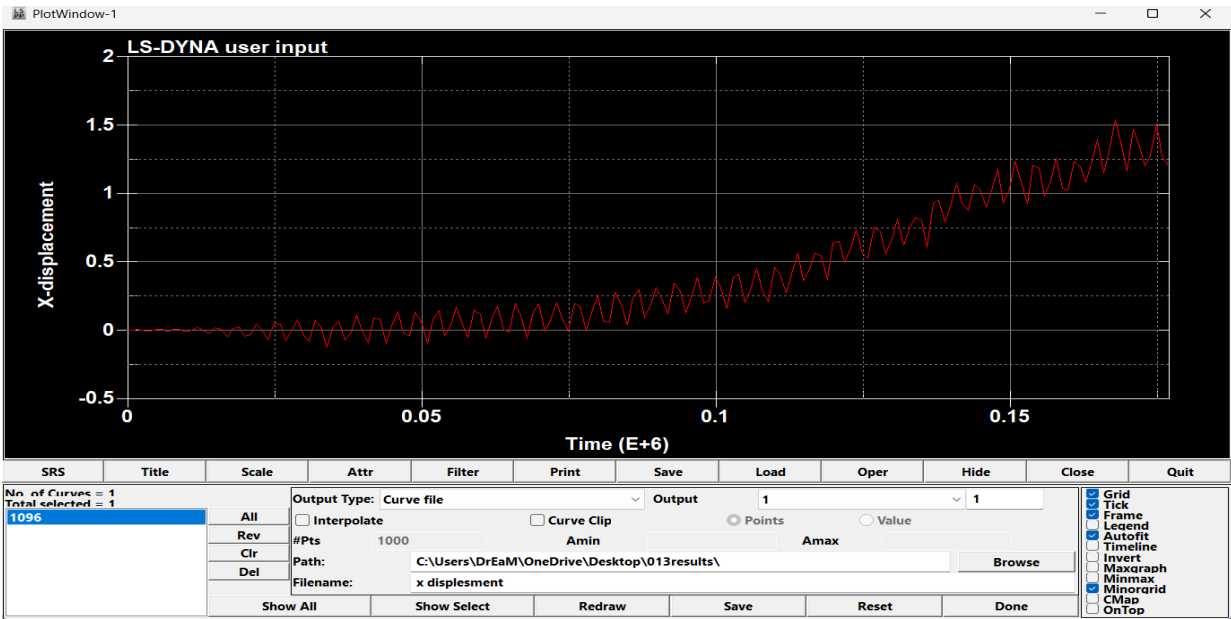
1-



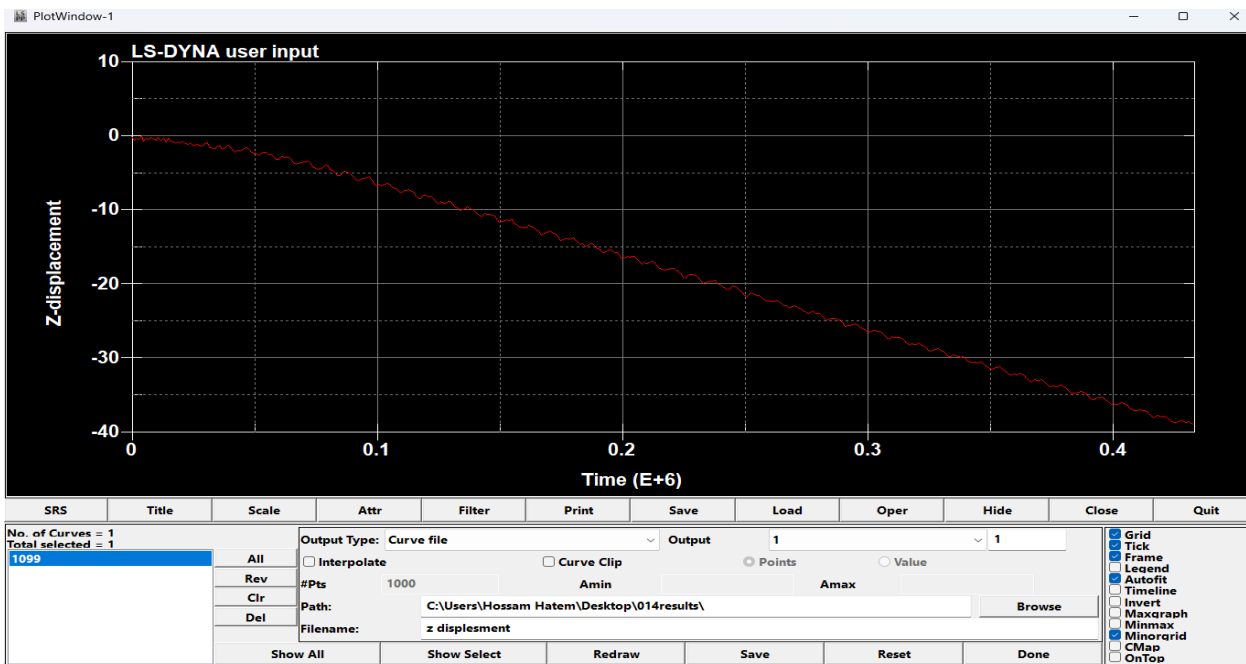
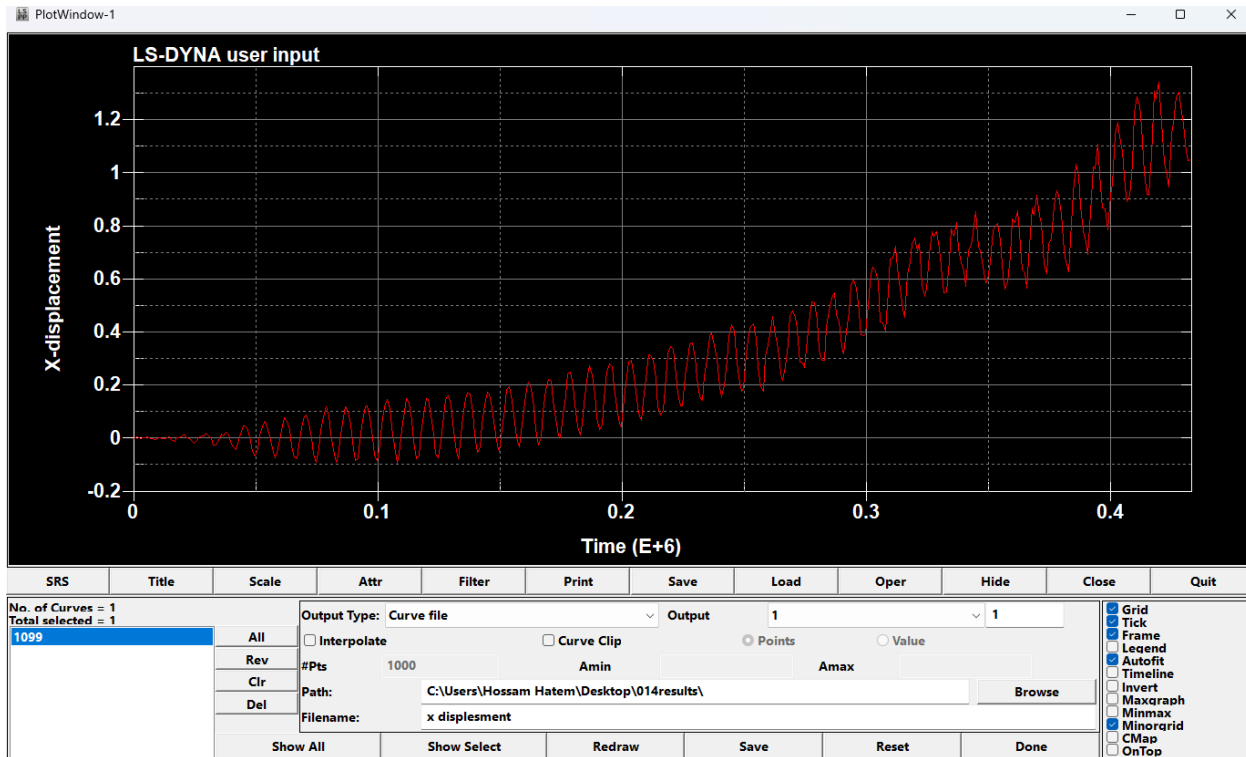
2-



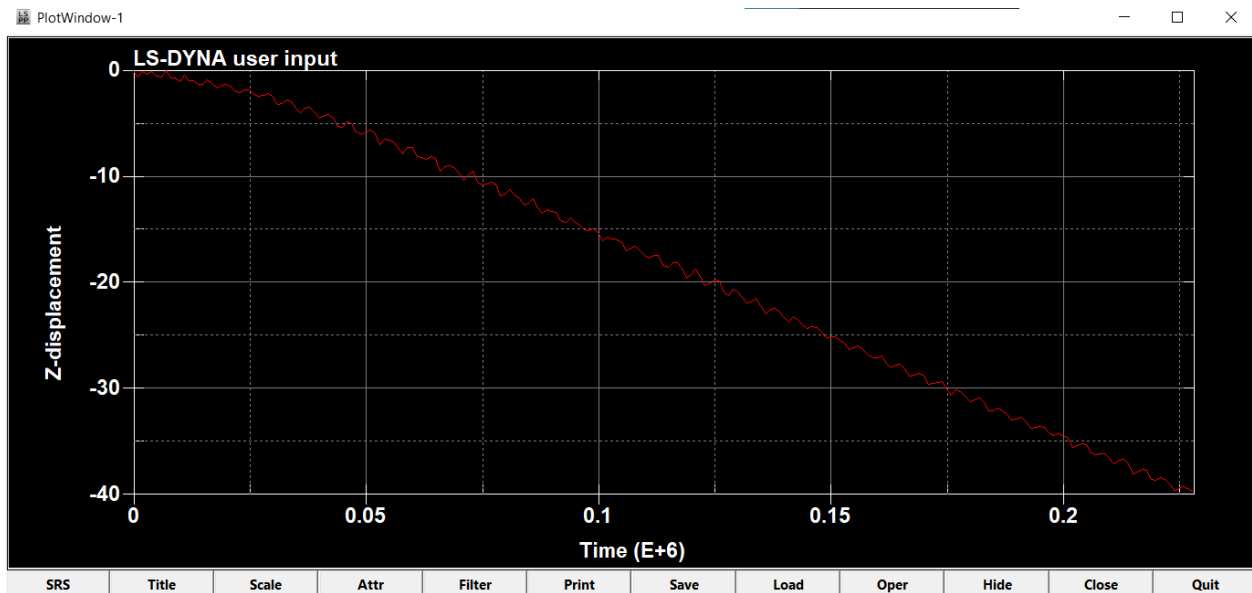
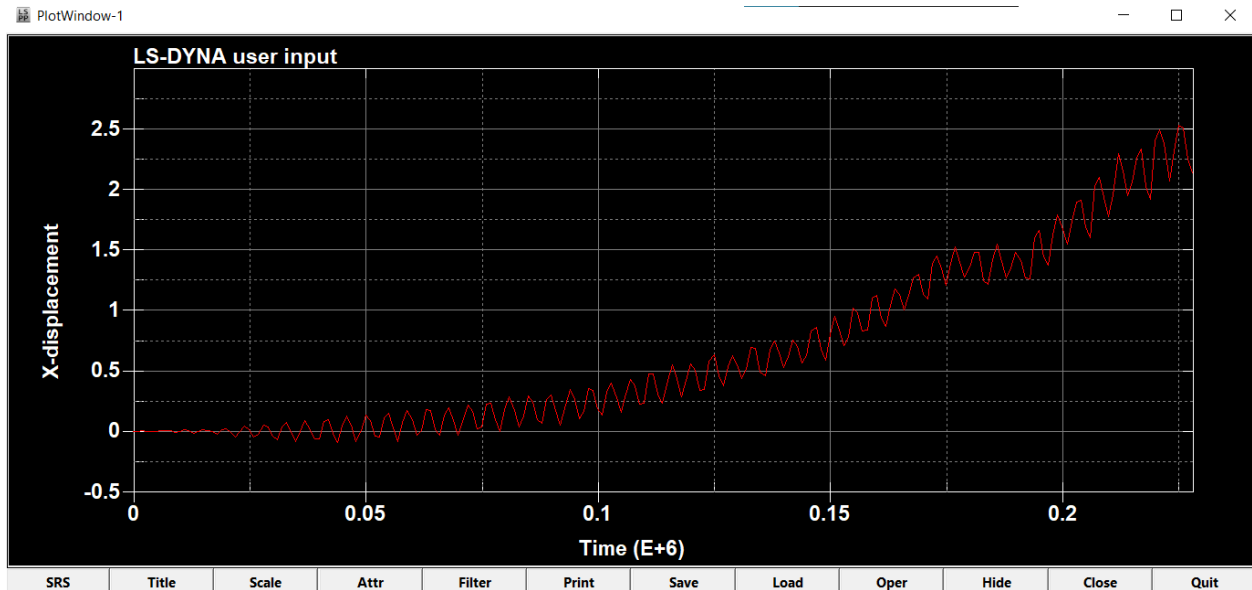
3-



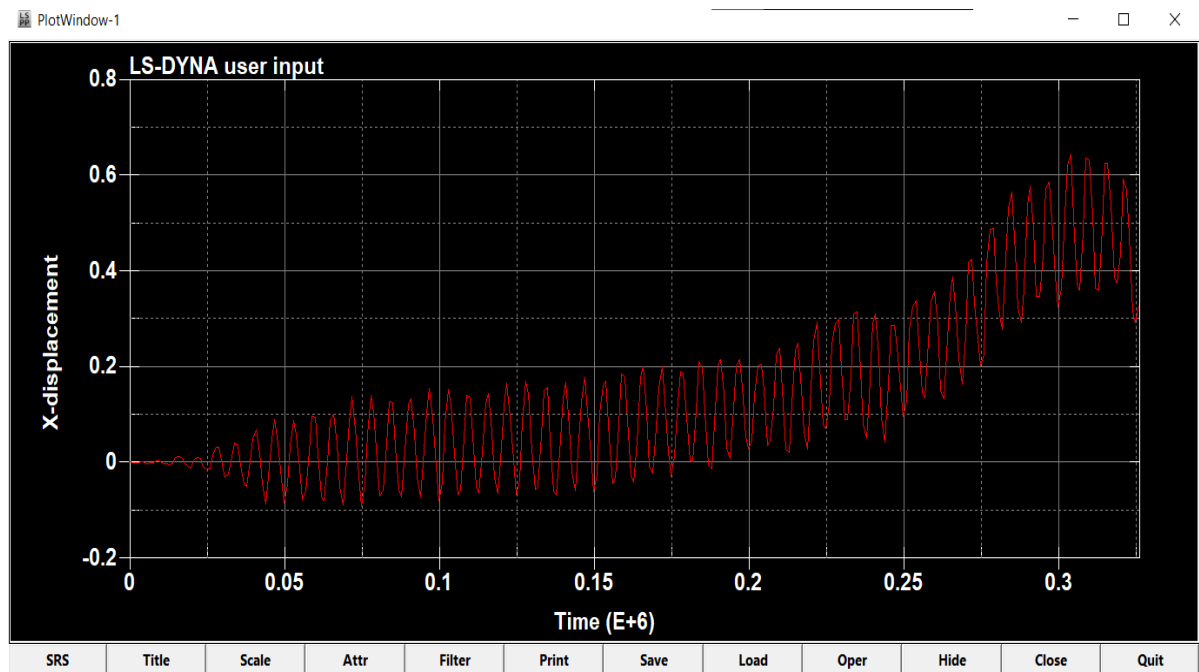
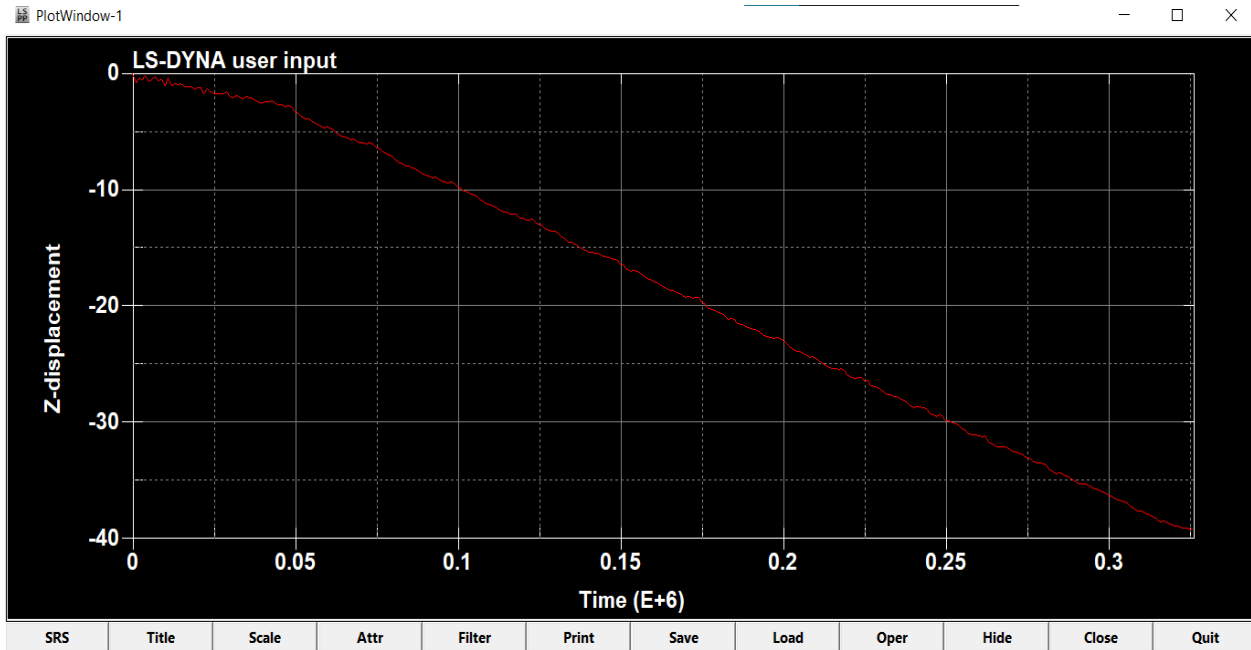
4-



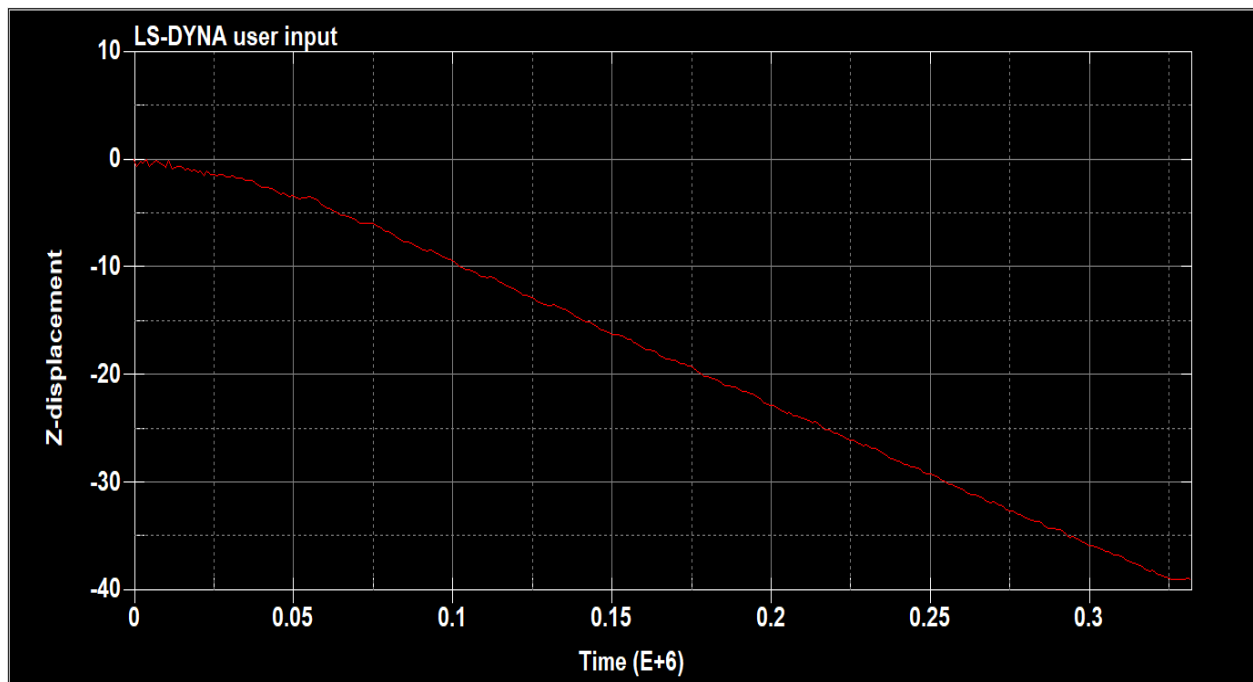
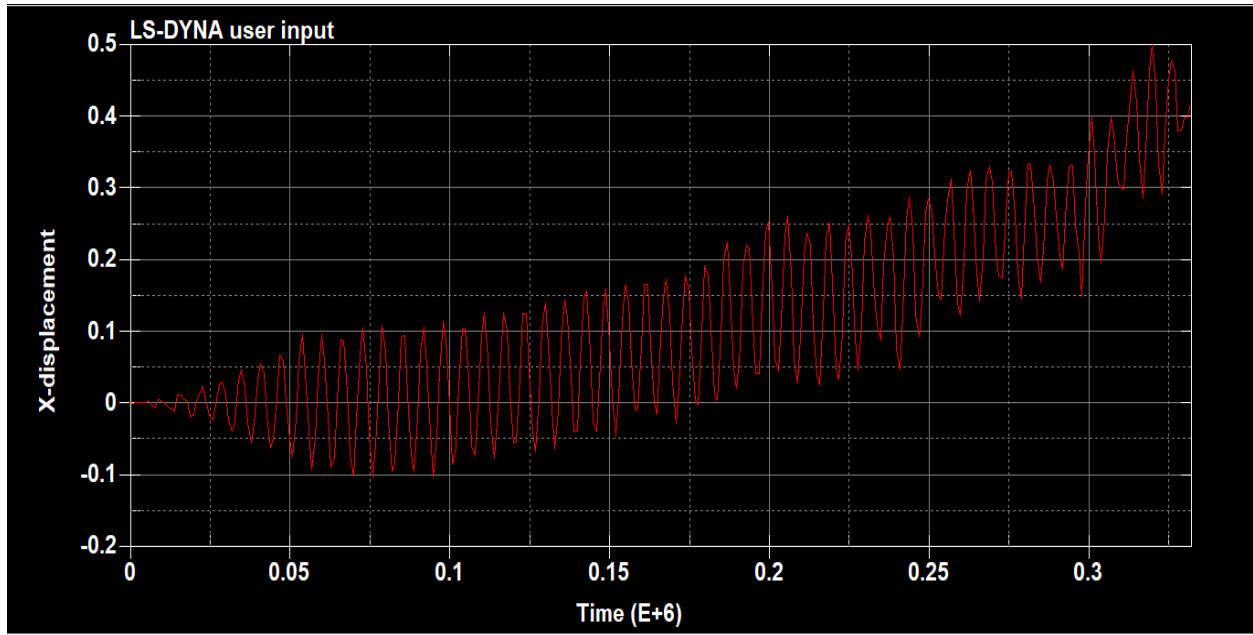
5-



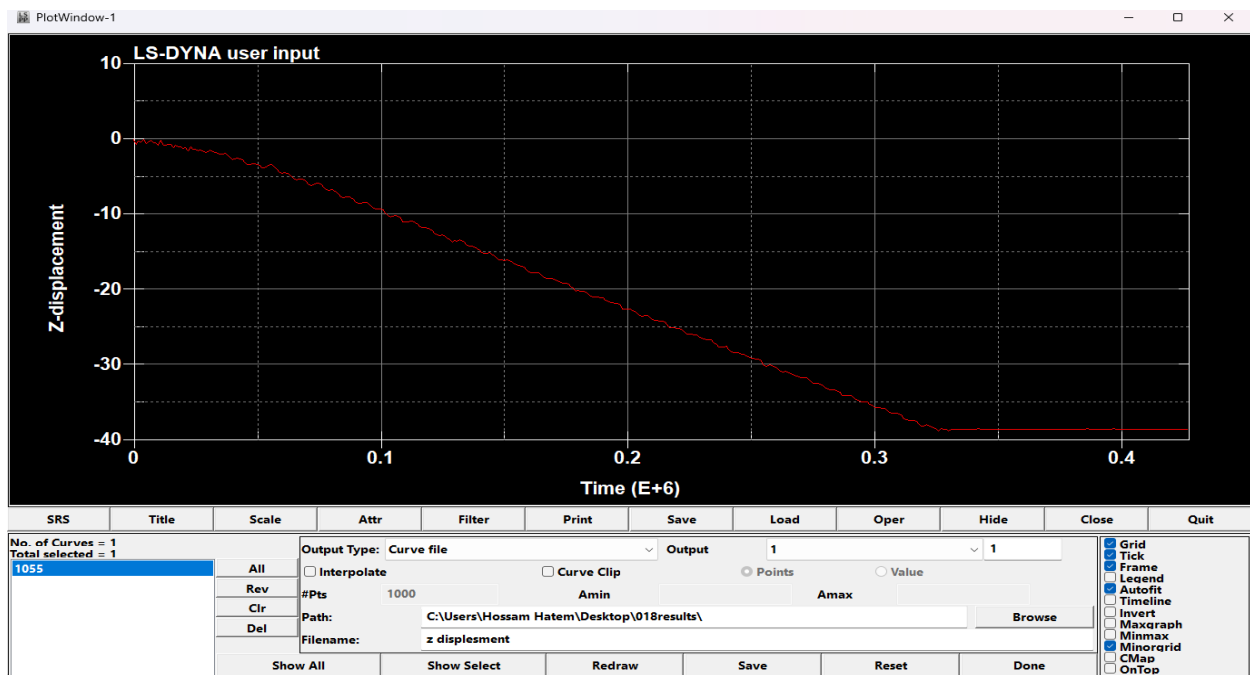
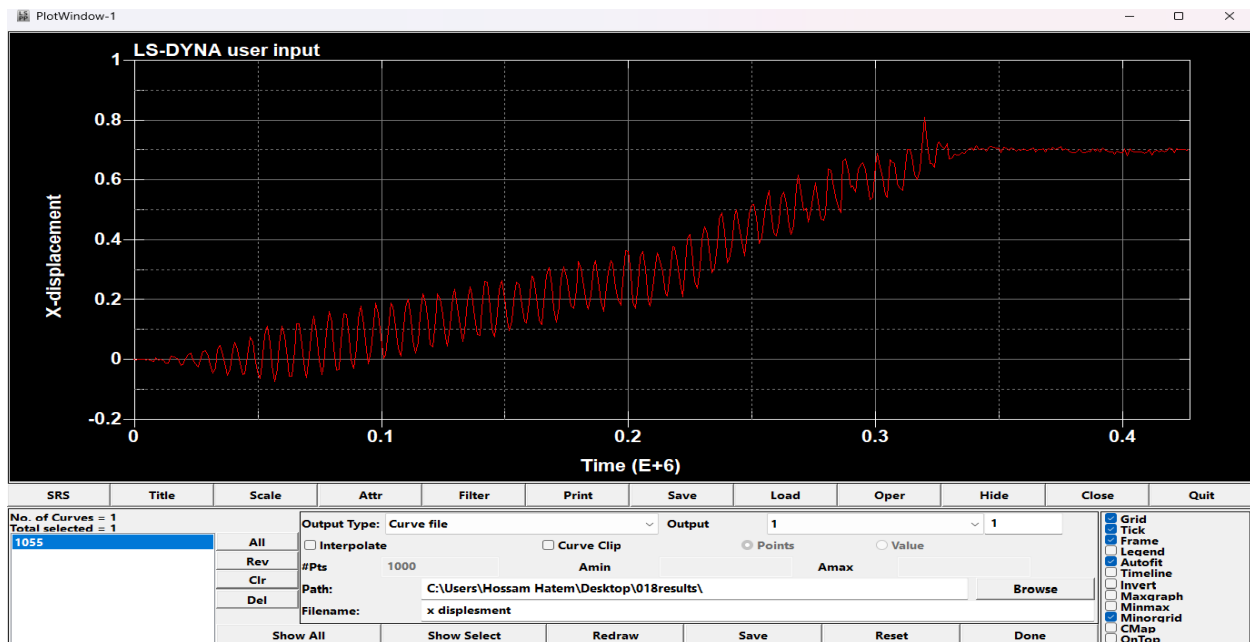
6-



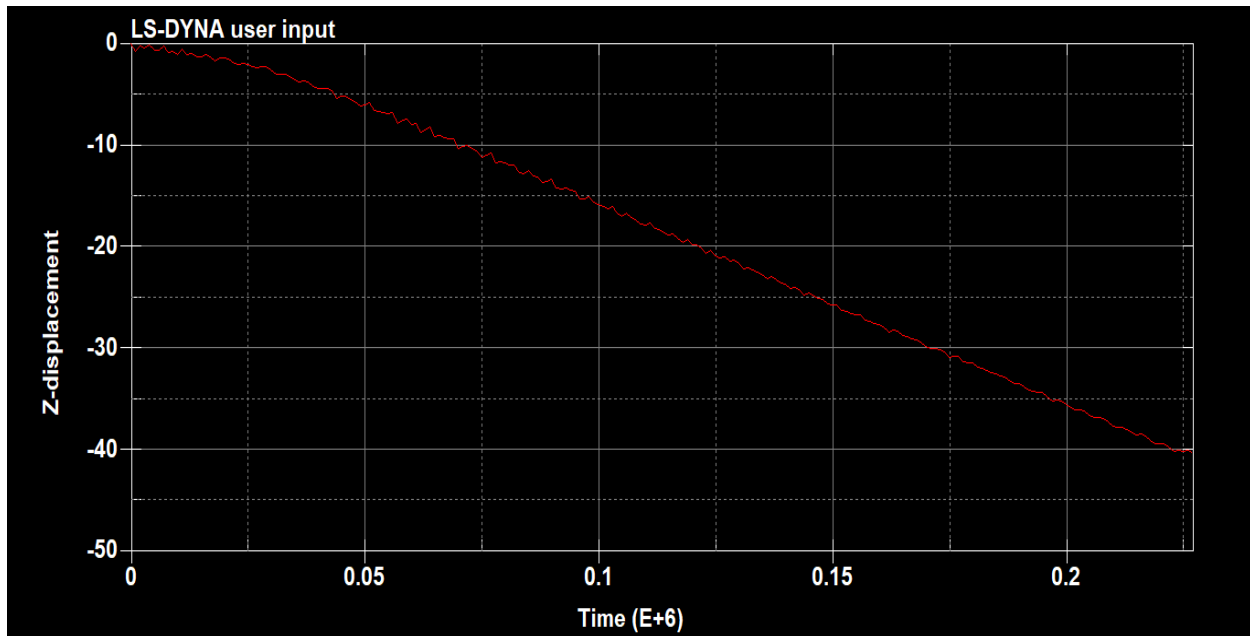
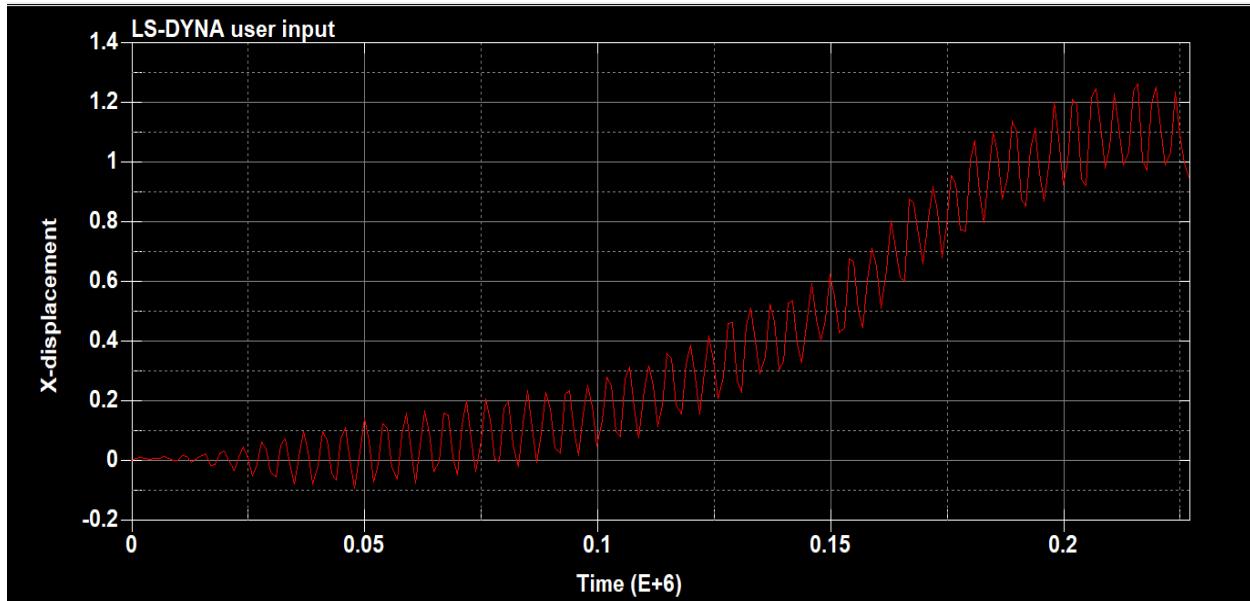
7-



8-



9-



Conclusion

This study explores the impact of various forming process parameters on the forming performance of incremental sheet metal forming (ISMF). The findings highlight several critical insights into the relationship between forming parameters and the resulting surface roughness, thickness reduction, and overall material behavior.

Surface Roughness and Sheet Thickness

The research revealed a significant correlation between the forming process parameters—spindle speed, feed rate, and incremental depth—and the surface roughness (Ra) and thickness reduction of the sheet metal. Higher spindle speeds and feed rates typically resulted in smoother surfaces but also led to greater thickness reduction. This trade-off underscores the necessity of optimizing these parameters to achieve the desired balance between surface finish and material integrity.

Stress Analysis

The Von Mises stress criterion was utilized to assess the stress distribution in the deformed sheet metal specimens. The study showed that the ISMF process induces varying stress levels across the specimens, influencing their deformation behavior. Understanding these stress distributions is crucial for predicting potential failure points and ensuring the safety and durability of the formed components.

Plastic Strain

Plastic strain measurements indicated that the material undergoes significant permanent deformation under the applied forces during the ISMF process. The extent of plastic strain varied among the specimens, influenced by the forming parameters. This information is vital for designing components that can withstand the required loads without permanent damage, enhancing the performance and reliability of the final products.

Shell Thickness

The effect of shell thickness on the forming performance was also examined. Thicker shells generally provided better structural integrity and reduced the likelihood of failure under load. However, they also posed challenges in terms of forming accuracy and surface quality. Therefore, careful consideration of shell thickness is essential in the design and manufacturing process to meet specific application requirements.

Displacement

The study analyzed the X and Z displacements of the sheet metal during the forming process. These displacements indicate the extent of deformation and the material's movement in the horizontal and vertical directions. Accurate control of these displacements is crucial for achieving the desired geometric accuracy and ensuring the dimensional stability of the formed components.

Recommendations for Future Work

1. **Parameter Optimization:** Further research should focus on developing advanced optimization algorithms to fine-tune the forming parameters for different materials and thicknesses, aiming to enhance both surface quality and material integrity.
2. **Material Characterization:** Comprehensive studies on different materials used in ISMF can provide deeper insights into their specific behavior under various forming conditions, leading to more tailored and efficient forming processes.
3. **Finite Element Analysis (FEA):** Expanding the use of FEA simulations can help predict the outcomes of the forming process more accurately, reducing the need for extensive experimental trials and accelerating the development of optimized forming strategies.
4. **Real-time Monitoring:** Implementing real-time monitoring systems during the forming process can provide immediate feedback on the forming conditions, allowing for dynamic adjustments to improve the process control and product quality.
5. **Sustainability:** Investigating the environmental impact of the ISMF process and exploring sustainable practices, such as material recycling and energy-efficient forming methods, can contribute to more eco-friendly manufacturing solutions.

Final Thoughts

The findings of this research contribute to a better understanding of the intricate relationships between forming process parameters and the performance of incremental sheet metal forming. By addressing the challenges associated with surface roughness, stress distribution, plastic strain, shell thickness, and displacement, this study provides a foundation for optimizing the ISMF process and enhancing the quality and reliability of formed components. Future research and technological advancements in this field will further drive the evolution of manufacturing techniques, paving the way for more efficient and sustainable production methods.

References:

1. Kalpakjian S, Schmid SR. Manufacturing engineering and technology, 2001. New Jersey: Prentice Hall. 2018.
2. Xu X. Integrating Advanced Computer-Aided Design, Manufacturing, and Numerical Control: Principles and Implementations: Principles and Implementations: IGI Global; 2009.
3. Altan T, Tekkaya AE. Sheet metal forming: processes and applications: ASM international; 2012.
4. Kumar Y, Kumar S, editors. Incremental sheet forming (ISF). Advances in Material Forming and Joining: 5th International and 26th All India Manufacturing Technology, Design and Research Conference, AIMTDR 2014; 2015: Springer.
5. Boljanovic V. Sheet metal forming processes and die design: Industrial Press Inc.; 2004.
6. Arfa H, Bahloul R, BelHadjSalah H. Finite element modelling and experimental investigation of single point incremental forming process of aluminum sheets: influence of process parameters on punch force monitoring and on mechanical and geometrical quality of parts. International journal of material forming. 2013;6:483-510.

7. Müller H, Enzmann H, editors. Potentials of rapid prototyping techniques for the manufacture of prototype sheet metal forming tools. Proc European Conf on Rapid Prototyping and Manufacturing, Aachen; 1998.
8. Uttarwar P, Raini S, Malwad D. Optimization of process parameter on Surface Roughness (Ra) and Wall Thickness on SPIF using Taguchi method. International Research Journal of Engineering and Technology. 2015;2(9):781-4.
9. Tisza M. General overview of sheet incremental forming. Journal of achievements in materials and Manufacturing Engineering. 2012;55(1):113-20.
10. Duflou JR, Habraken A-M, Cao J, Malhotra R, Bambach M, Adams D, et al. Single point incremental forming: state-of-the-art and prospects. International Journal Of Material Forming. 2018;11(6):743-73.
11. Ebeling H, Fink H, Luo M, Geus H. Cellulose meltblown nonwovens using the lyocell-process. Lenzinger Berichte. 2006;86:124-31.
12. Zhao X, Ou H. A new flexible multi-point incremental sheet forming process with multi-layer sheets. Journal of Materials Processing Technology. 2023;322:118214.
13. Önal Ü, Seçgin Ö, Özsert İ, Çoban A, Cesur İ. A New Approach to Multi-Stage Incremental Forming Method. Arabian Journal for Science and Engineering. 2024:1-10.
14. Lee H-R, Lee M-G, Park N. Effect of evolutionary anisotropic hardening on the prediction of deformation and forming load in incremental sheet forming simulation. Thin-Walled Structures. 2023;193:111231.
15. Coman CC, Mazurchevici S-N, Carausu C, Nedelcu D. Analysis of Incremental Sheet Forming of Aluminum Alloy. Materials. 2023;16(19):6371.
16. Doss QM, Abaas TF, Bedan AS. The effect of tool path strategy on twist behavior in Single Point Incremental Sheet Metal Forming. Journal of Engineering, Baghdad University. 2013;4.
17. Resende TP, Carmo GP, Afonso DG, Alves de Sousa RJ. Development of a Compact Incremental Forming Machine. Machines. 2024;12(2):86.

- Kumar A, Gulati V, Kumar P. Investigation of surface roughness in incremental sheet forming. *Procedia computer science*. 2018;133:1014-20. 18.
- Zhu H, Ou H. Heat-assisted friction stir incremental sheet forming of thermoplastics. *Journal of Materials Processing Technology*. 2024;328:118396. 19.
- Seçgin Ö, Özsert İ. Optimization of Titanium Grade 2 sheet forming by roller blank holder method. *Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering*. 2023;237(6):2585-95. 20.
- Mohan SR, Dewang Y, Sharma V, editors. Tool path planning for hole-flanging process using single point incremental forming. *MATEC Web of Conferences*; 2024: EDP Sciences. 21.
- Kumar Y, Kumar S, editors. Incremental sheet forming (ISF). *Advances in Material Forming and Joining: 5th International and 26th All India Manufacturing Technology, Design and Research Conference, AIMTDR 2014*; 2015: Springer. 22.
- Li J, Li S, Xie Z, Wang W. Numerical simulation of incremental sheet forming based on GTN damage model. *The International Journal of Advanced Manufacturing Technology*. 2015;81:2053-65. 23.
- Li W, Attallah MM, Essa K. Heat-assisted incremental sheet forming for high-strength materials—a review. *The International Journal of Advanced Manufacturing Technology*. 2023;124(7):2011-36 24.

